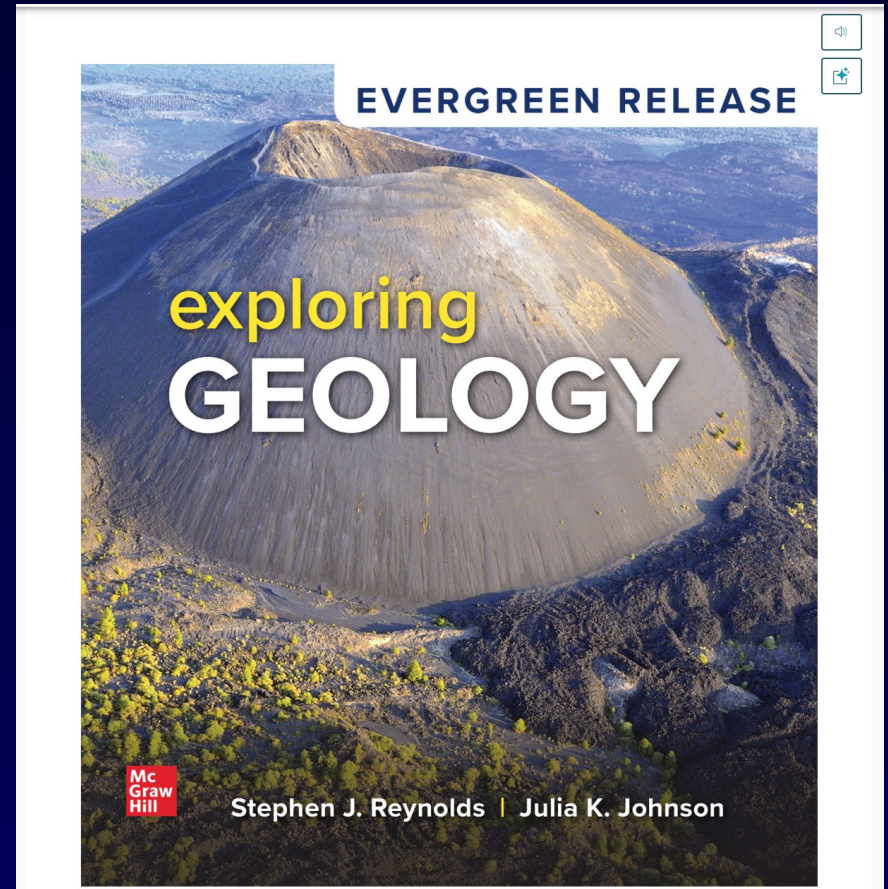
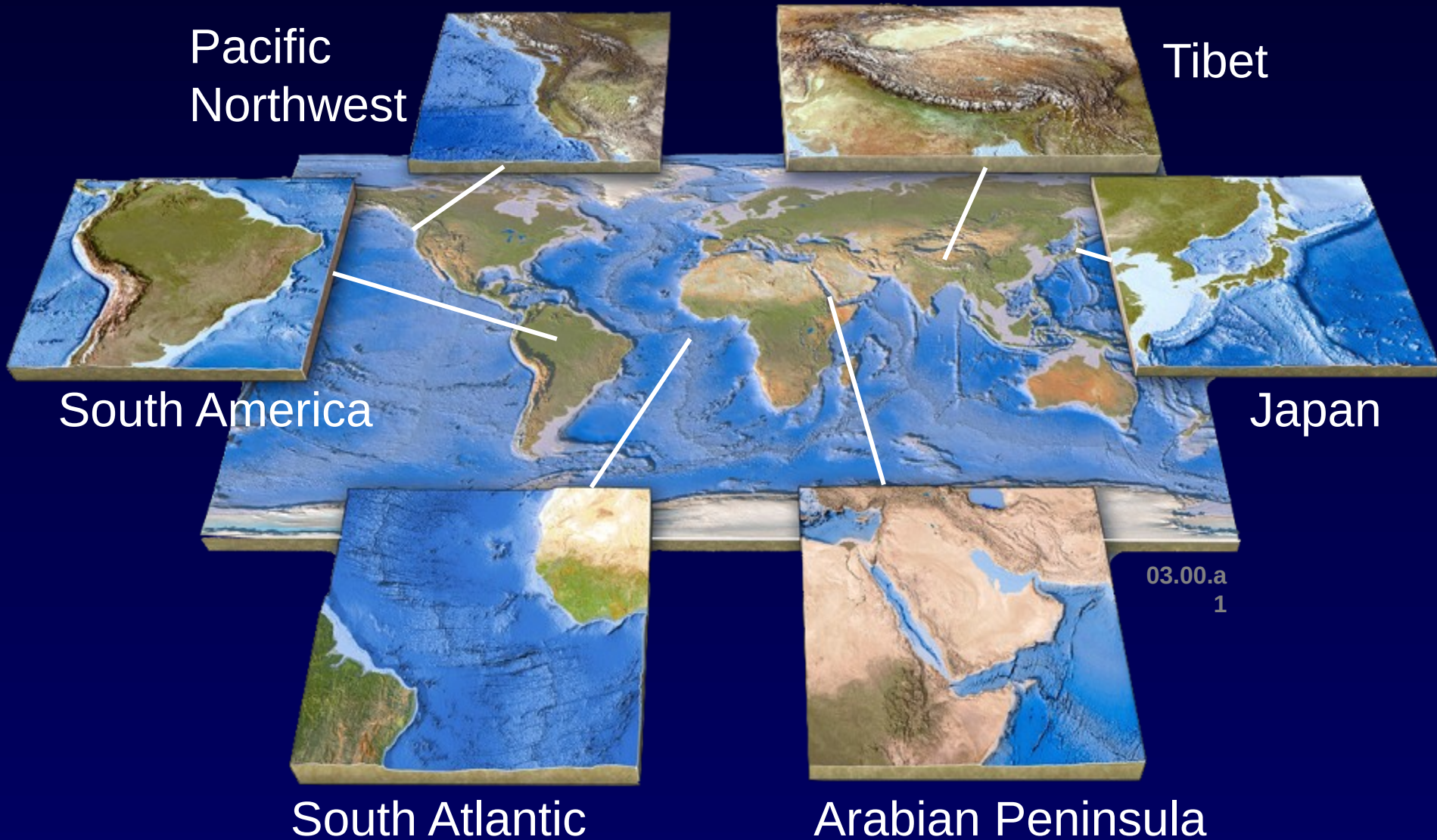


PLATE TECTONICS

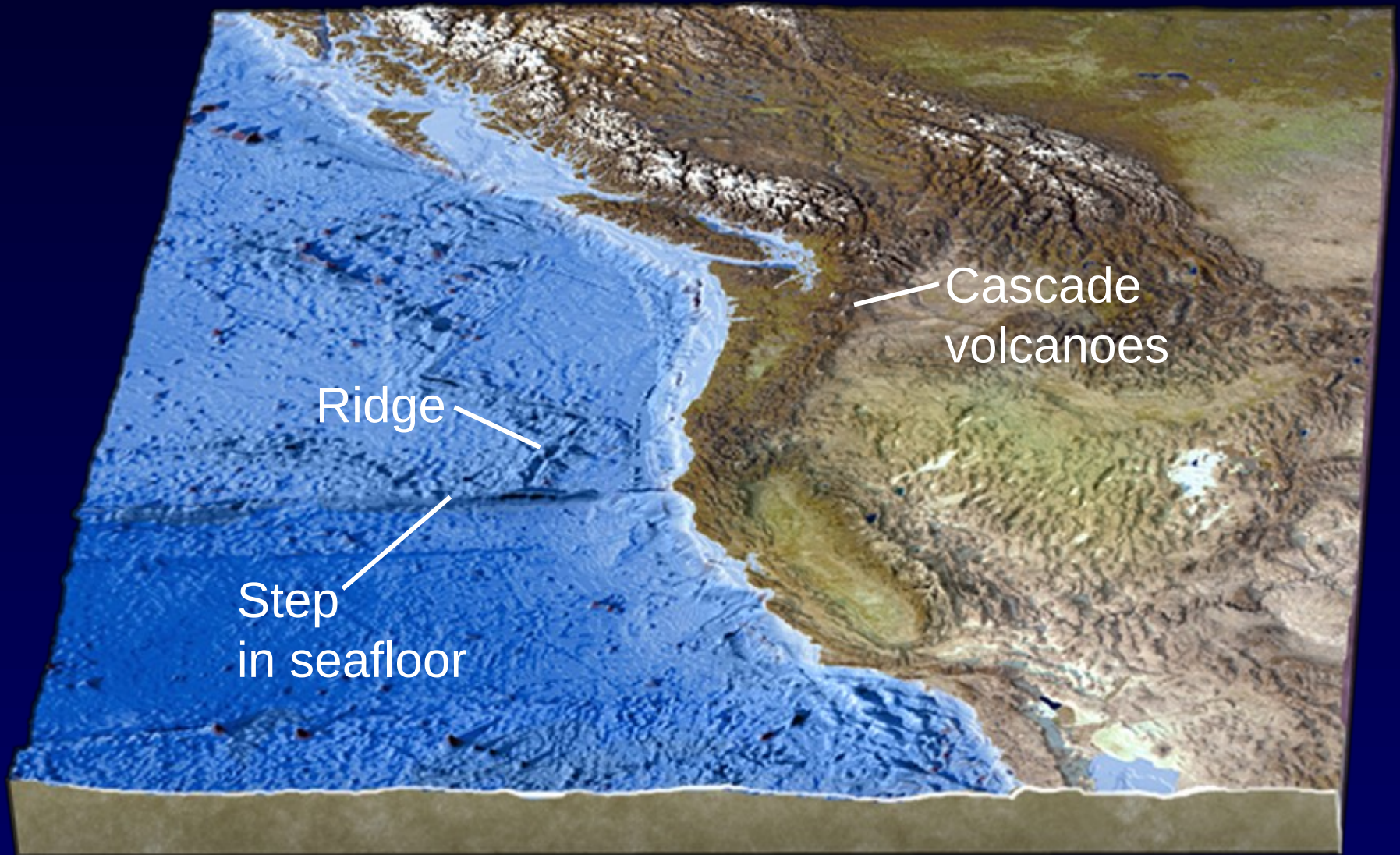
CHAPTER 3



Observe the world map and each enlargement

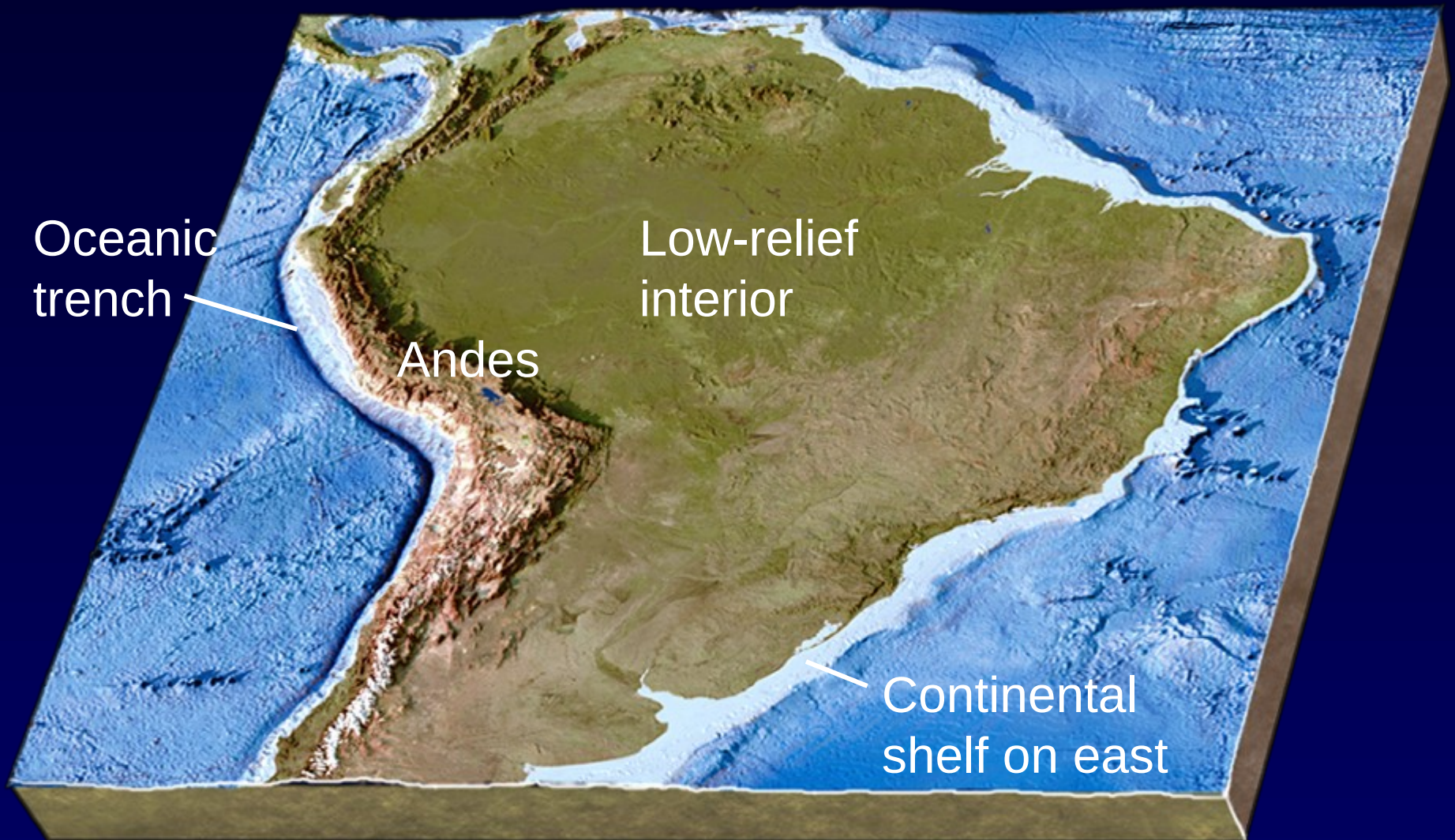


Observe features near the Pacific Northwest



03.00.a1a

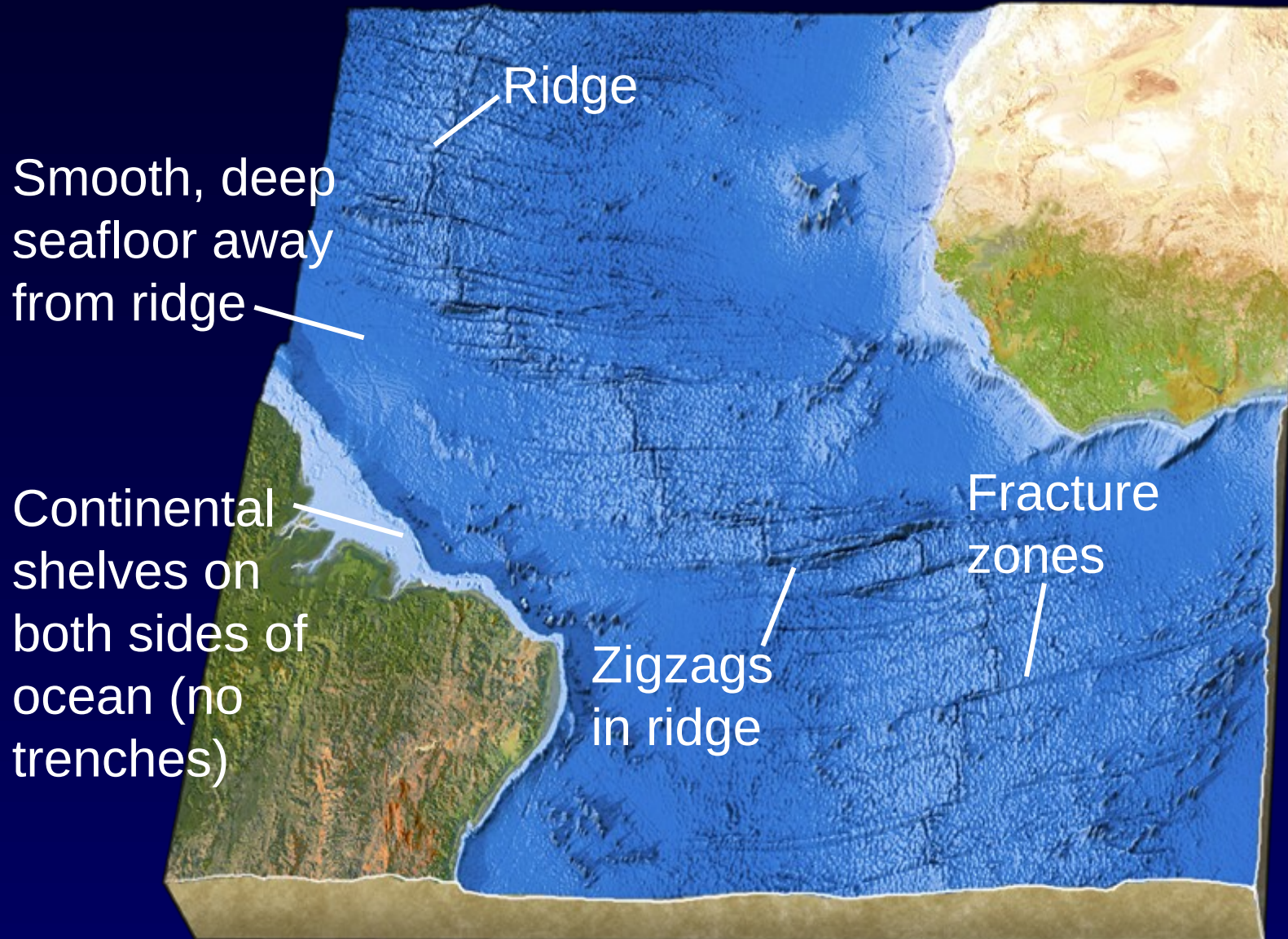
Observe features near South America



03.00.a1b



Observe features of the South Atlantic Ocean



03.00.a1c

Observe features near the Tibetan Plateau



Tibetan Plateau

Himalaya —

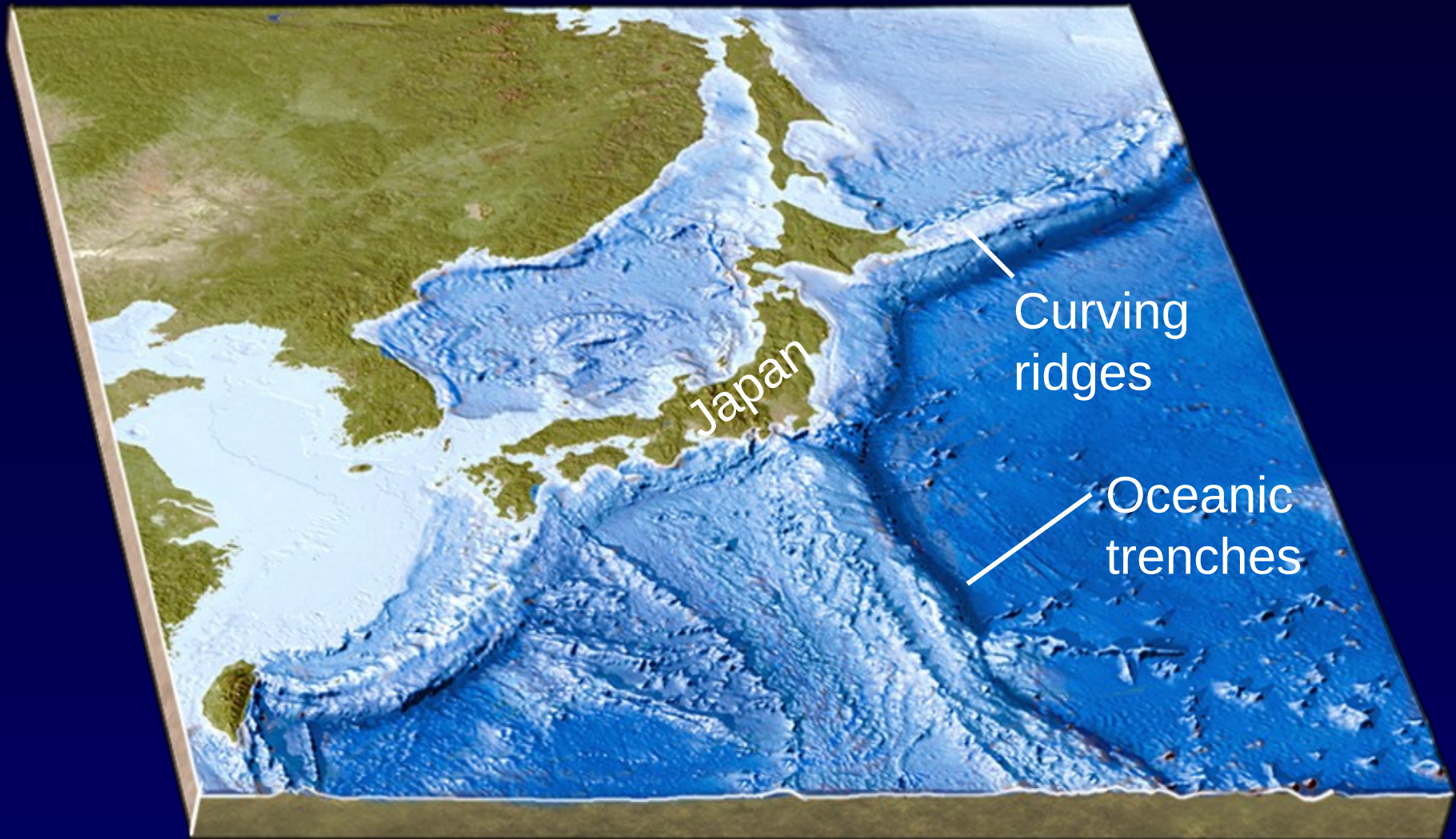
Lowlands
in India

03.00.a1d

The 14 eight-thousanders

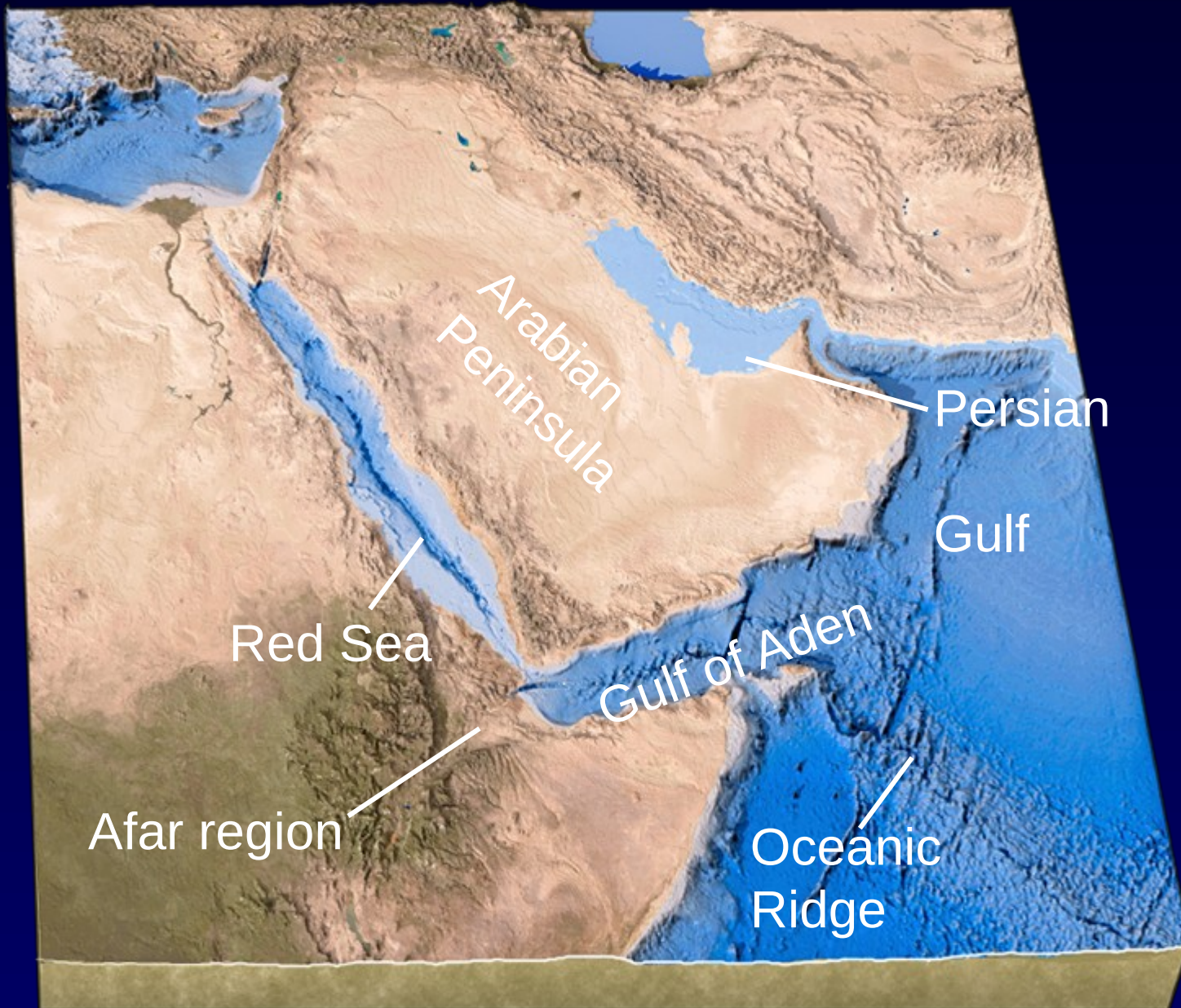


Observe features near Japan



03.00.a1e

Observe features near the Arabian Peninsula



Observe some of the main features on Earth's surface

Oceanic fracture zones

Continental shelves

Submerged
ridges and
island arcs

Plateaus

Linear island
and seamount
chains

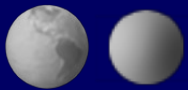
Deep ocean
trenches

Continents

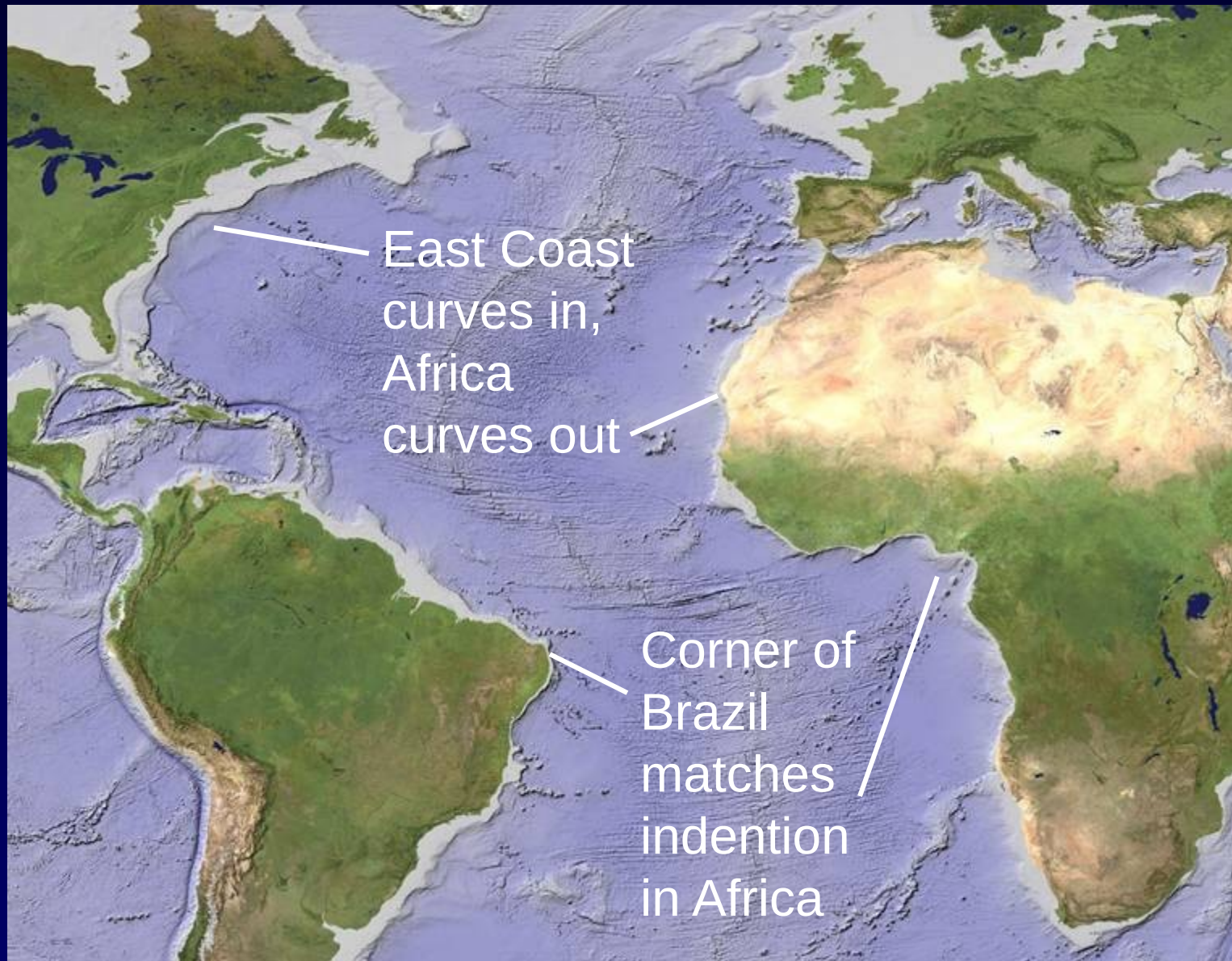
Mid-ocean ridges

Oceanic plateaus

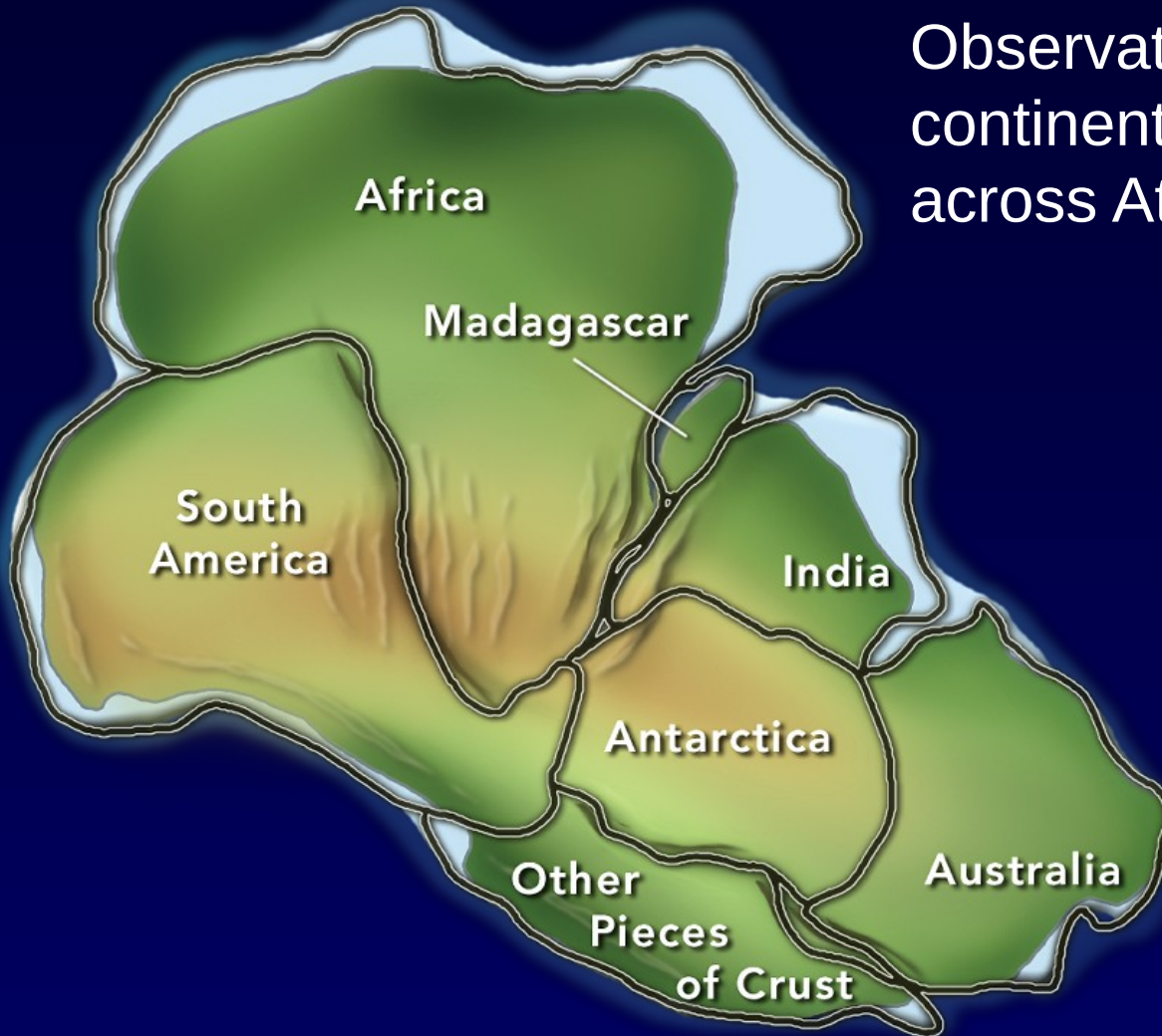
03.01.a
1



Compare the shapes of the continental edges on opposite sides of the Atlantic Ocean



Some Continents Have Shapes that Fit Together



Observation: Shapes of continents match across Atlantic Ocean

Interpretation: Continents fit back together because once joined and later moved apart (Continental Drift hypothesis)

Fossil Data and Continental Drift



Observation: Fossils of land creatures found on continents now separated by oceans

Interpretation: Continents were joined so creatures could walk from one place to another

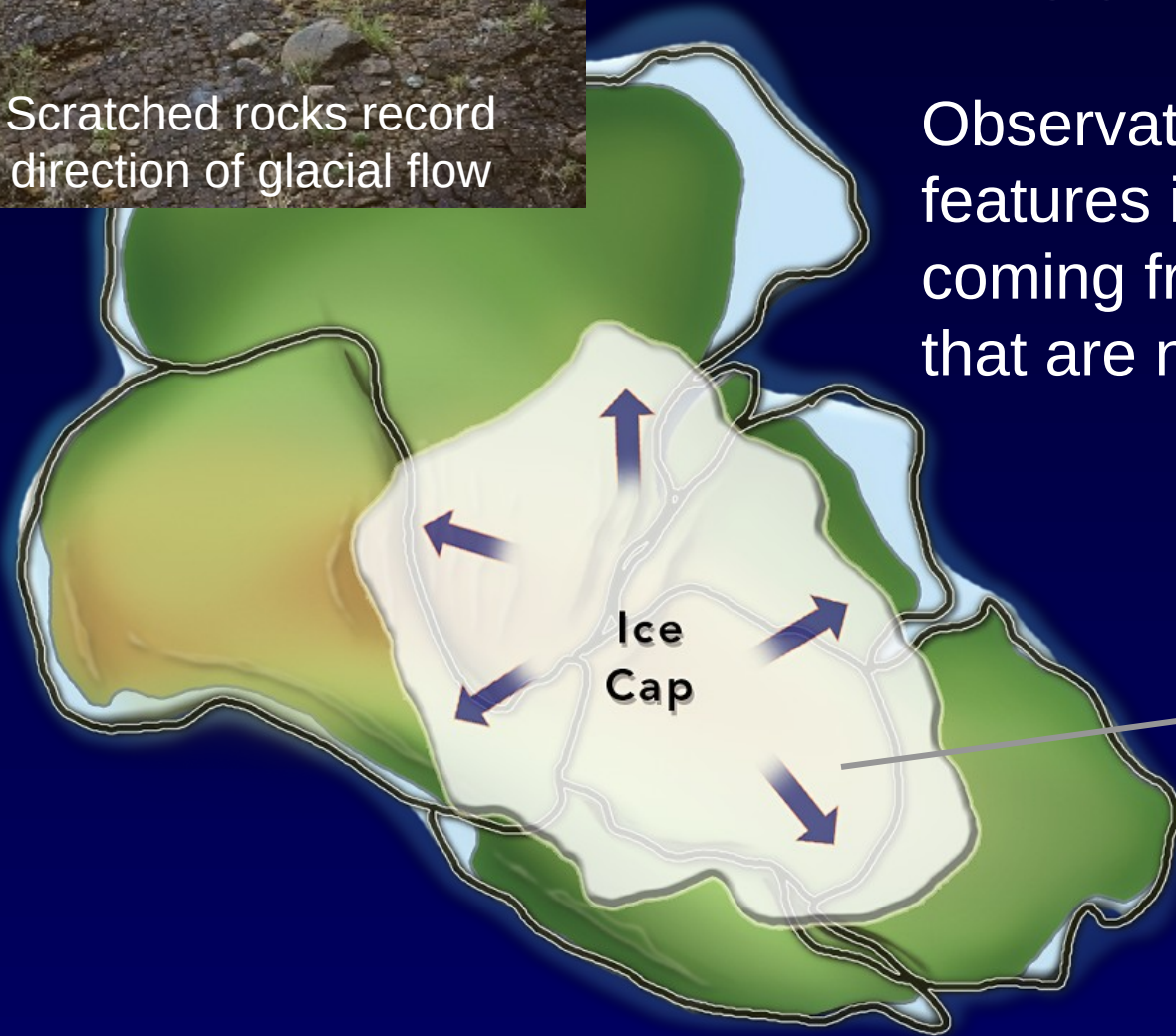
Glaciers and Continental Drift



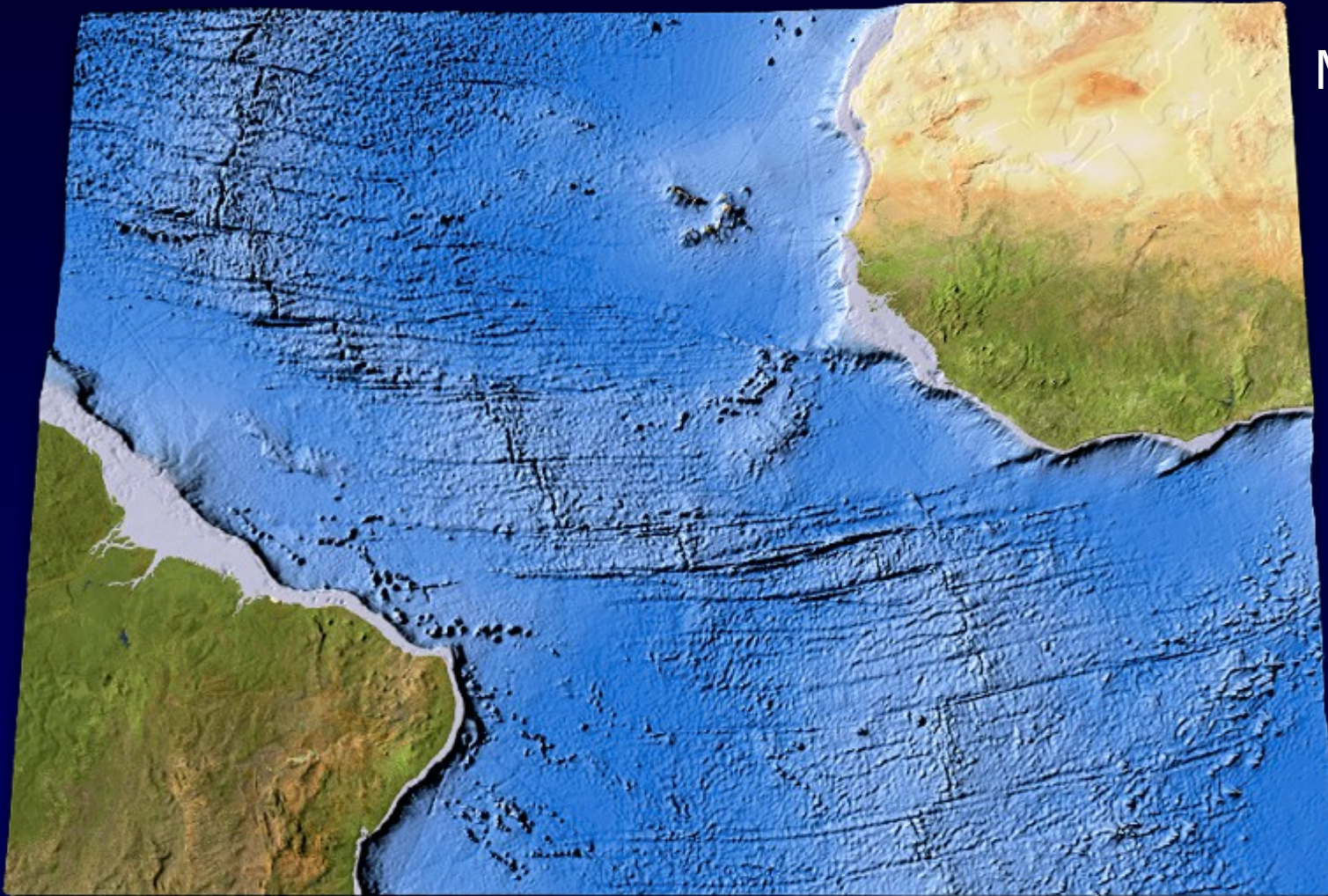
Scratched rocks record direction of glacial flow

Observation: Glacial features indicate glaciers coming from directions that are now oceans

Interpretation: When continents restored, ice sheets moved in predictable directions, away from center



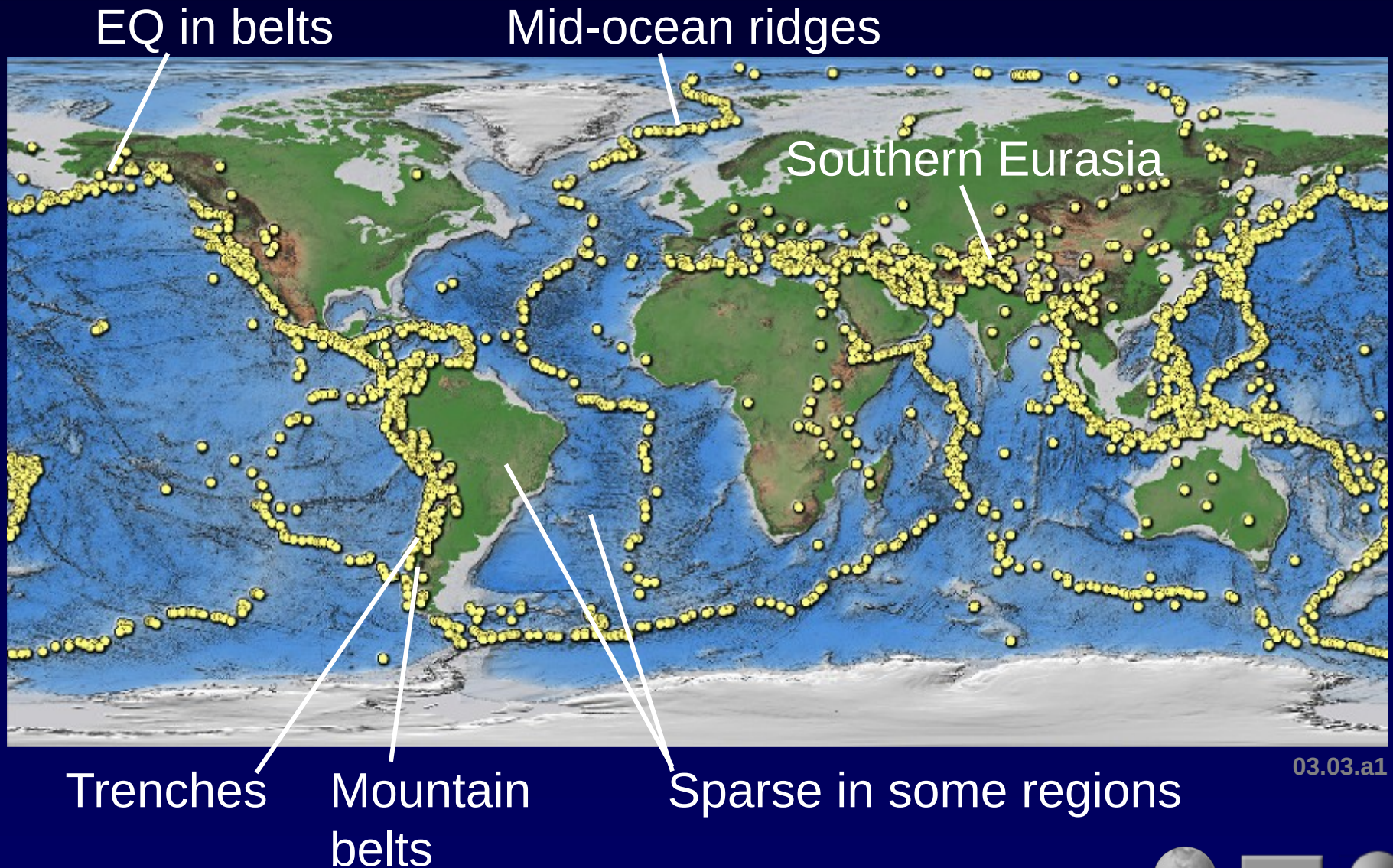
Problems with Continental Drift



Mechanism

Some geologists' lack of familiarity with data from other parts of world

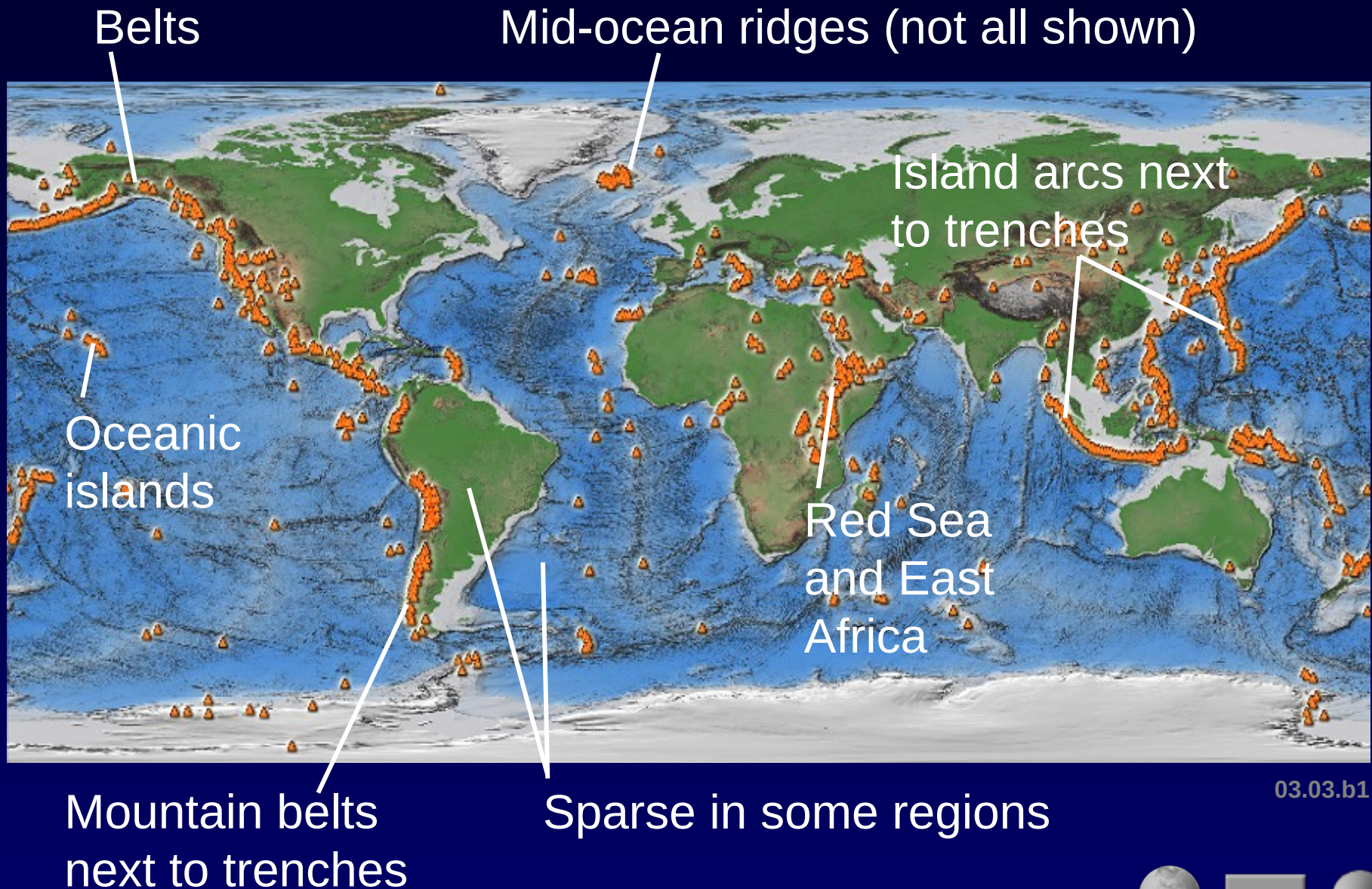
Observe the pattern of earthquakes (yellow dots)



03.03.a1



Observe the pattern of volcanoes (orange triangles)



03.03.b1



Observe the pattern of elevations (brown = high)

Mountains in belts

Southern Eurasia high

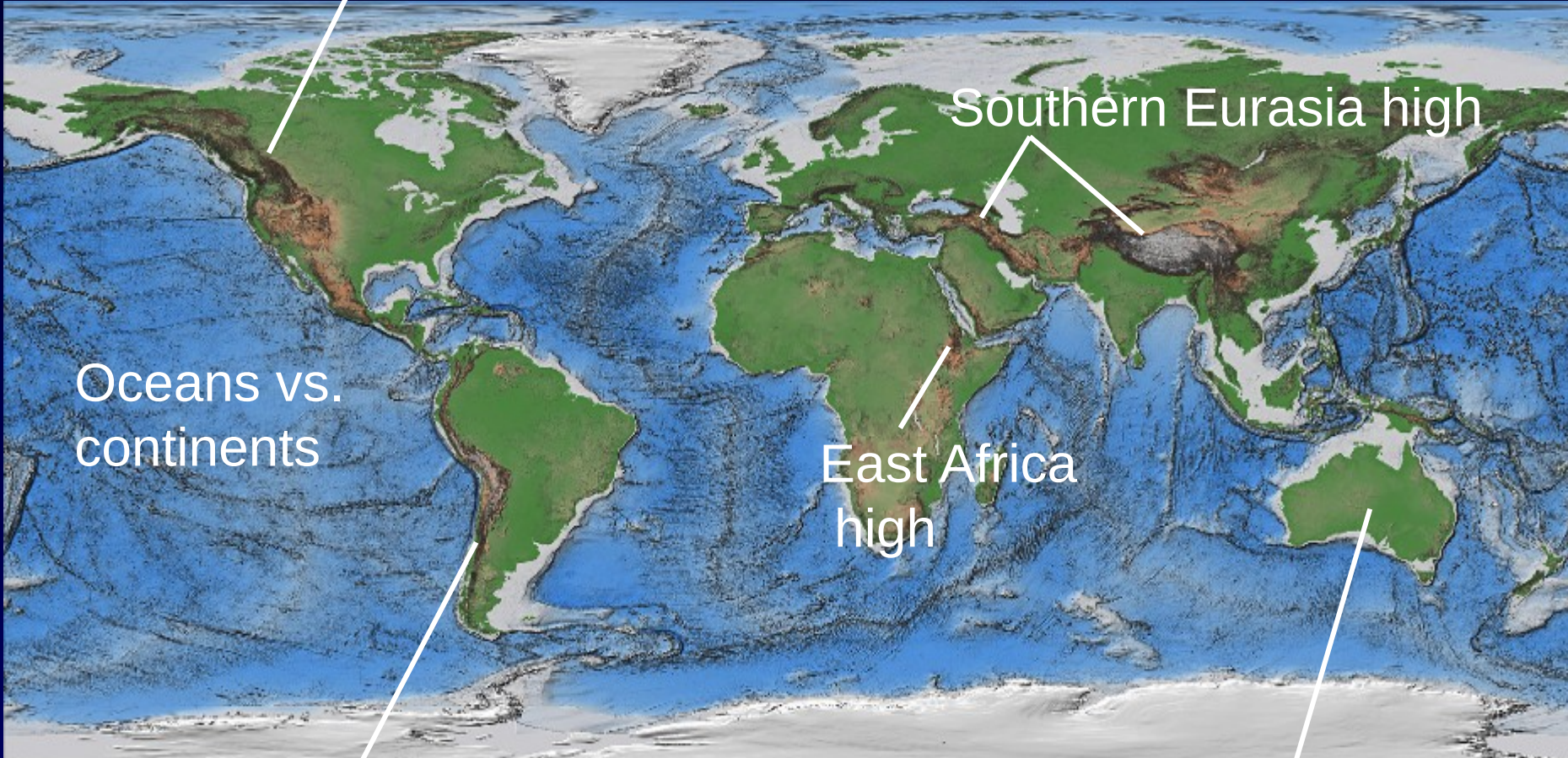
Oceans vs.
continents

East Africa
high

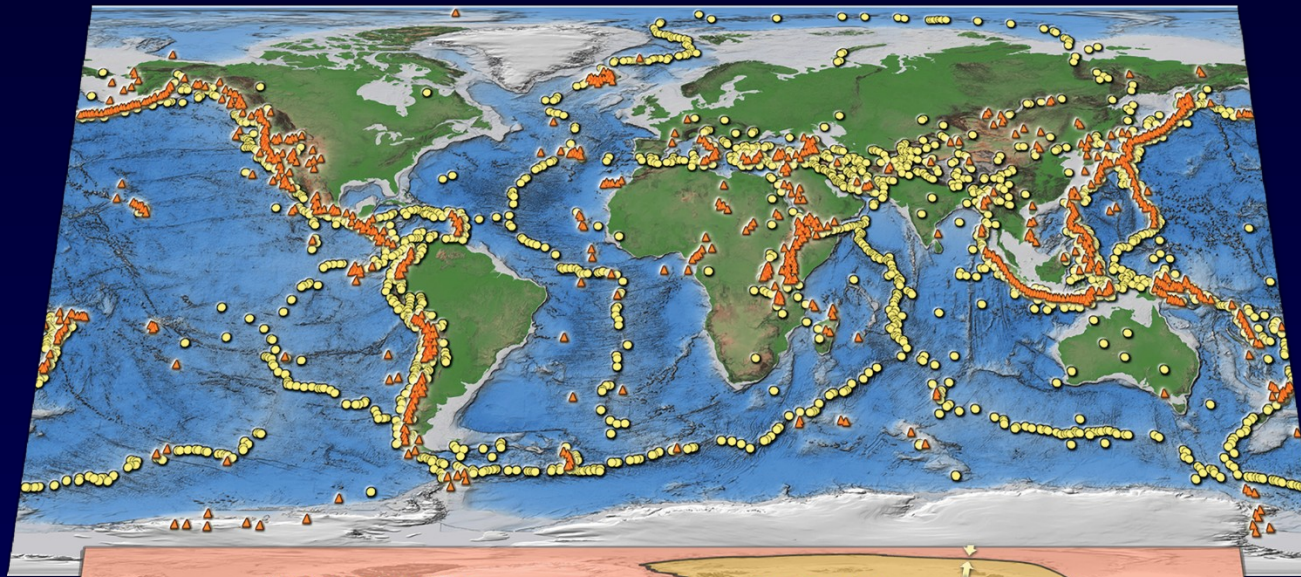
Some mountain belts
next to trenches

Most parts of continents
gentle and low

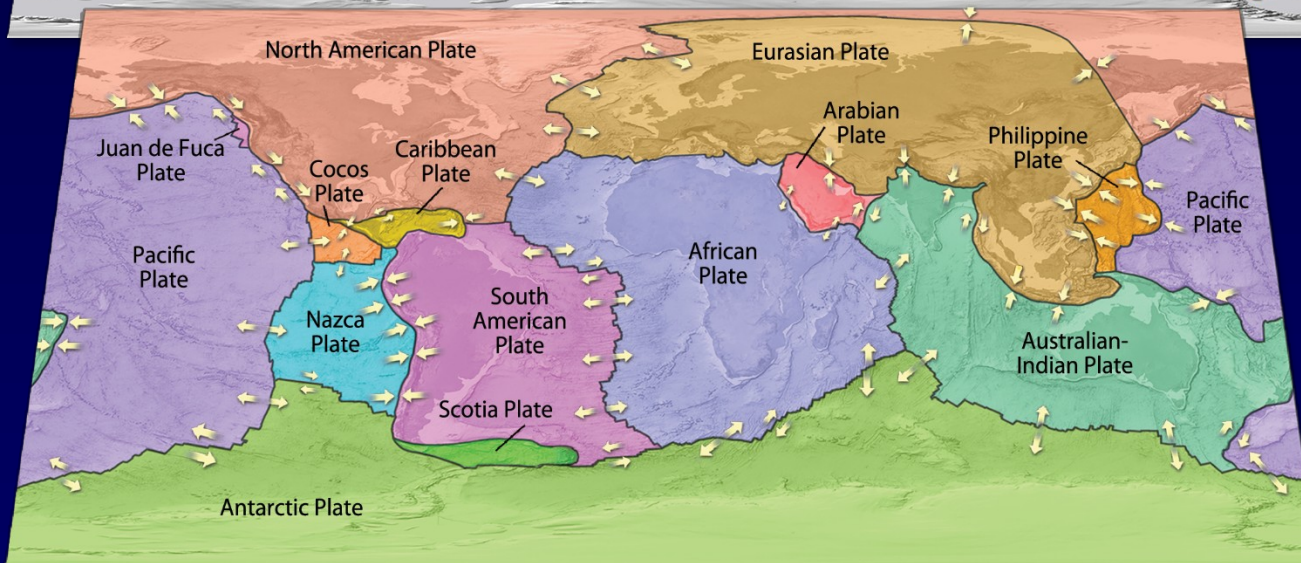
03.03.b2



Compare the distribution of earthquakes (yellow), volcanoes (orange), and high elevations (brown)



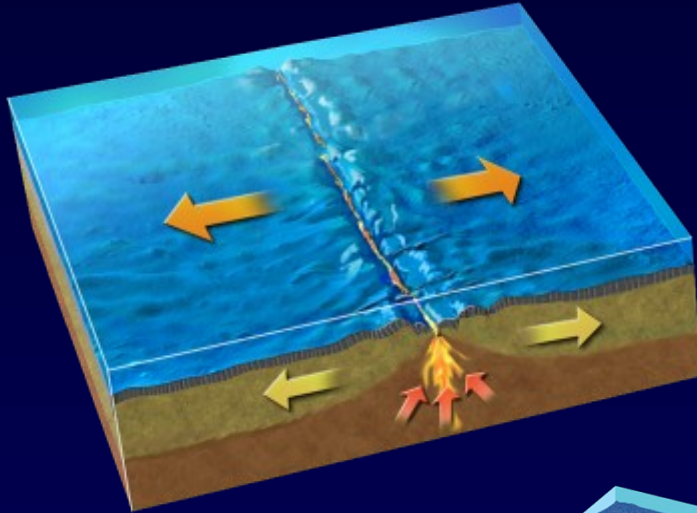
EQ, volcanism, or mountain building = *tectonic activity*



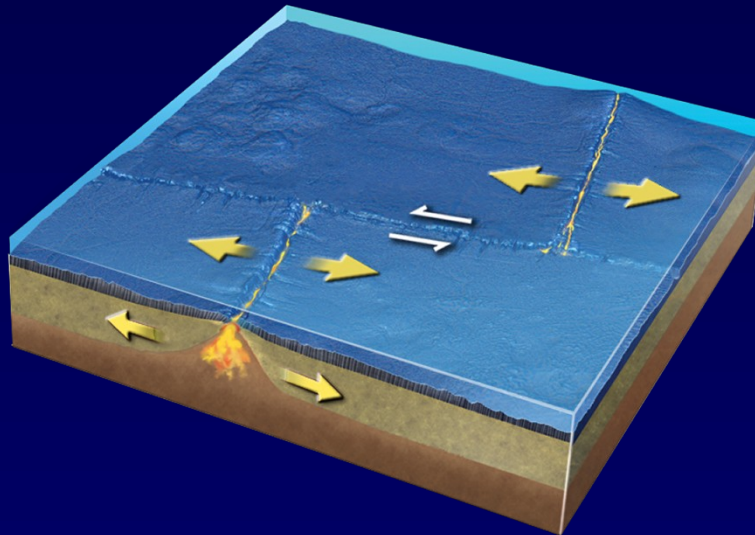
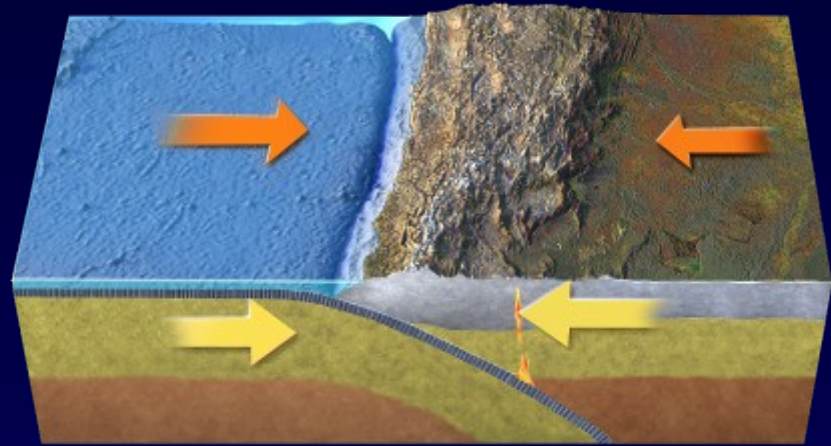
Belts of tectonic activity divide lithosphere into *tectonic plates*

Three Types of Relative Plate Motions

Move apart:
divergent boundary



Move toward each other:
convergent boundary



Move horizontally
past one another:
transform boundary

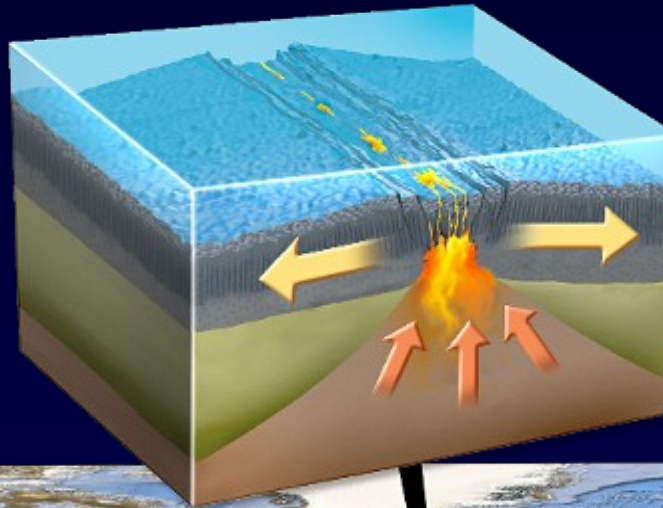
/ Divergent / Convergent / Transform

Compare these locations with mid-ocean ridges, oceanic trenches, mountains belts, lines of islands, etc.



Oceanic Divergent Boundaries

Oceanic plates
move apart at
mid-ocean ridges
(seafloor
spreading)



Forms new
oceanic crust



Features and Processes of Mid-Ocean Ridges

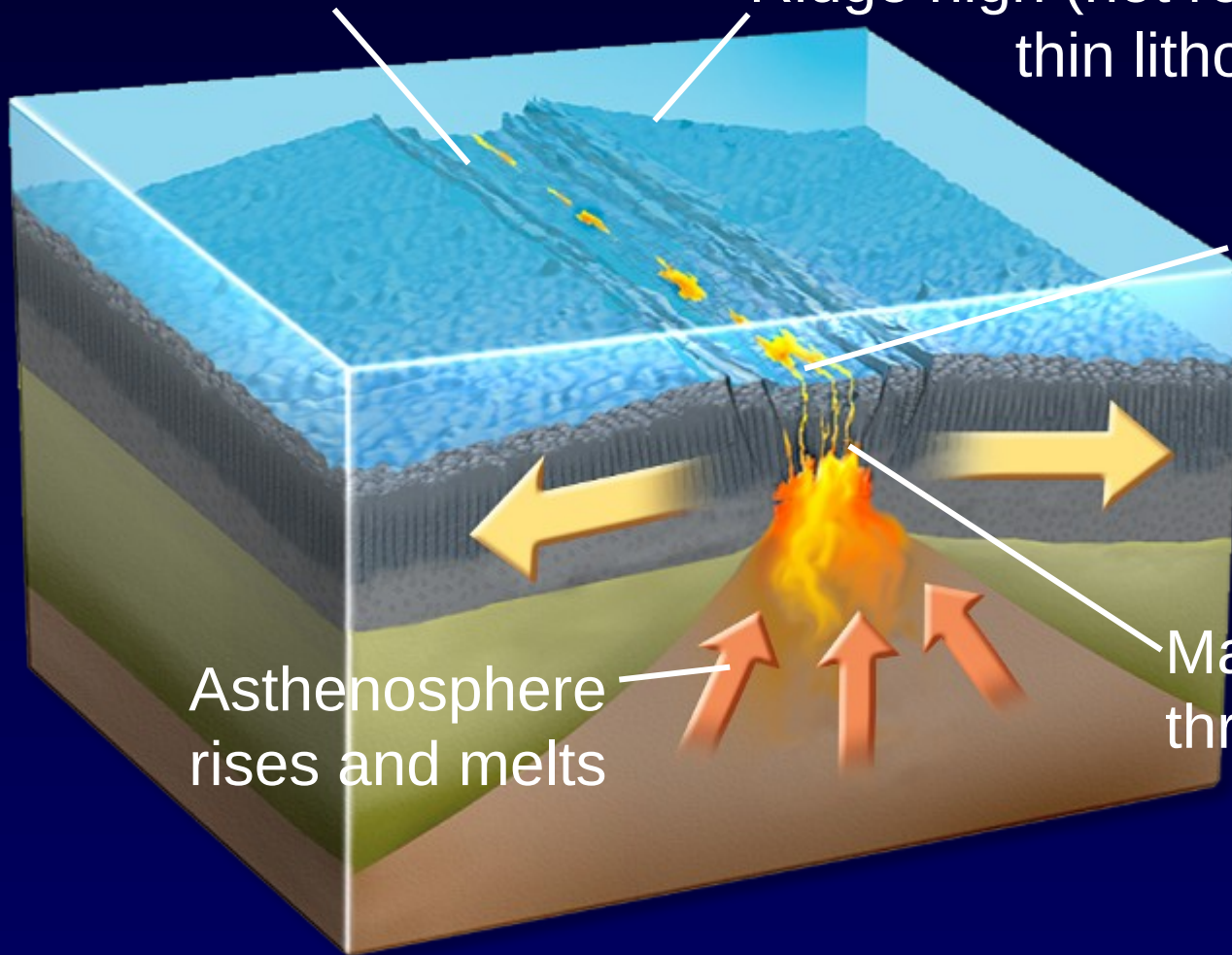
Narrow trough, or *rift*

Ridge high (hot rocks and thin lithosphere)

Magma erupts or solidifies at depth; forms new oceanic crust

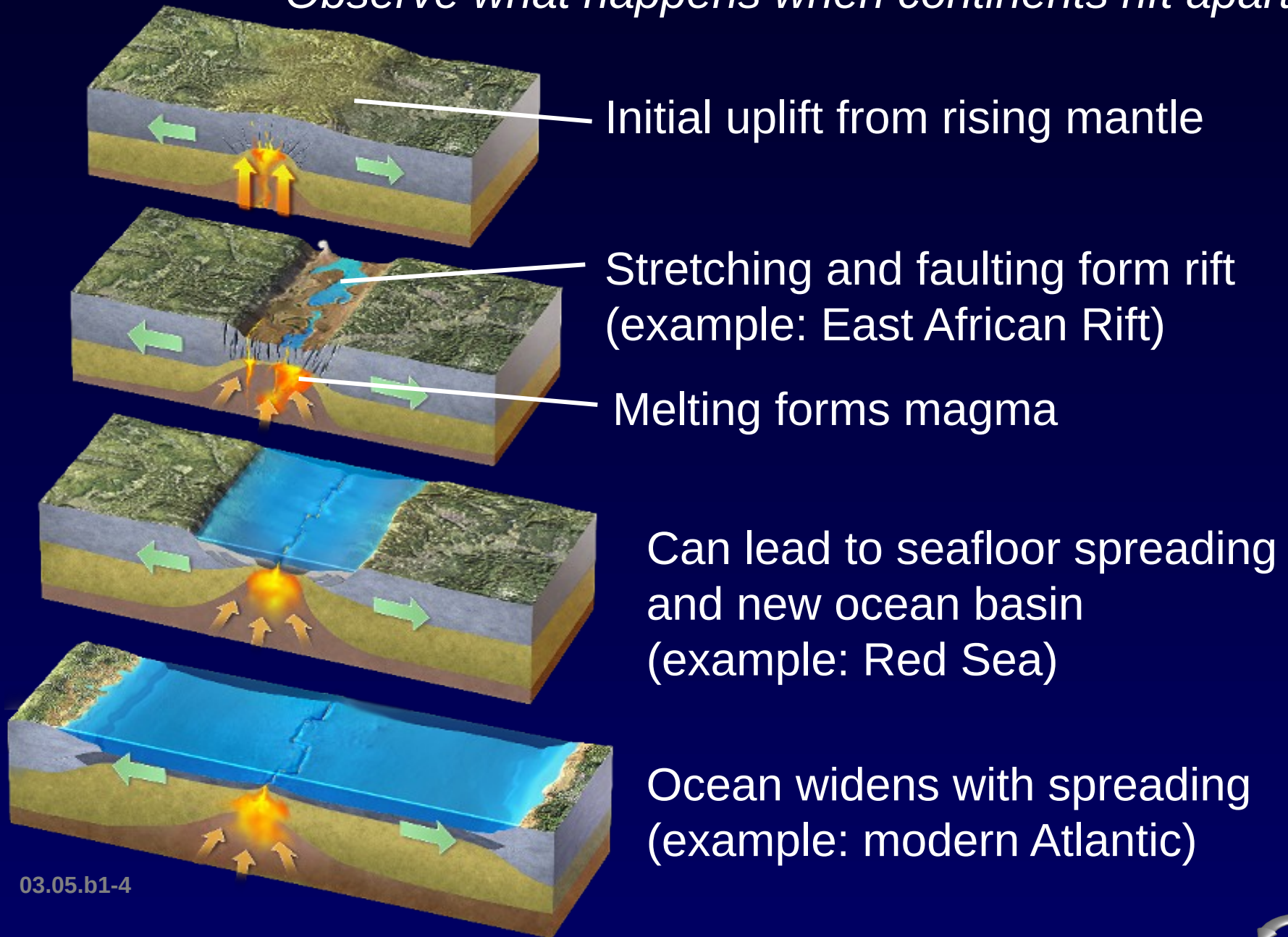
Magma rises through fractures

Asthenosphere rises and melts



Sketch a mid-ocean ridge, labeling the processes in your own words

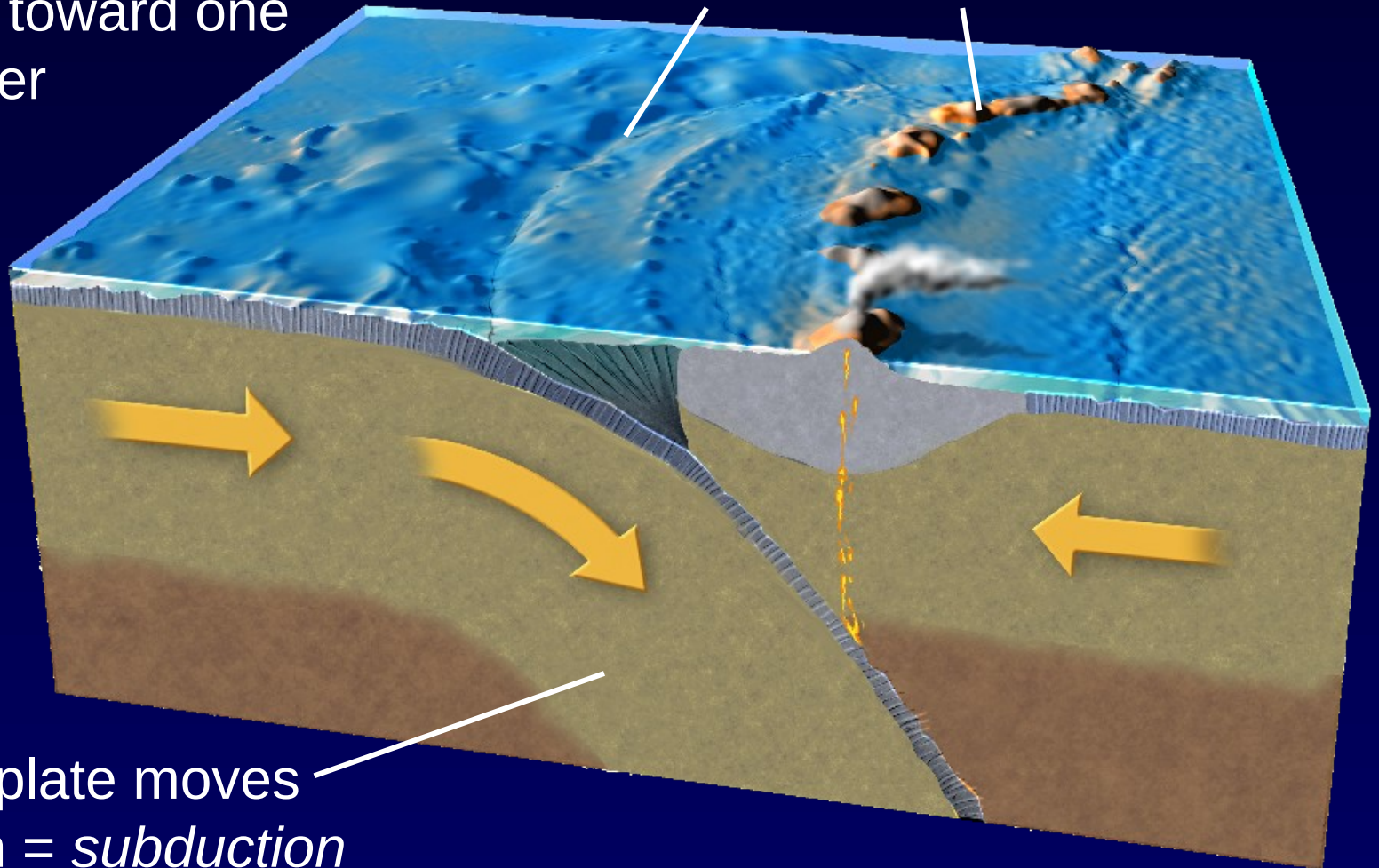
Observe what happens when continents rift apart



Ocean-Ocean Convergent Boundary

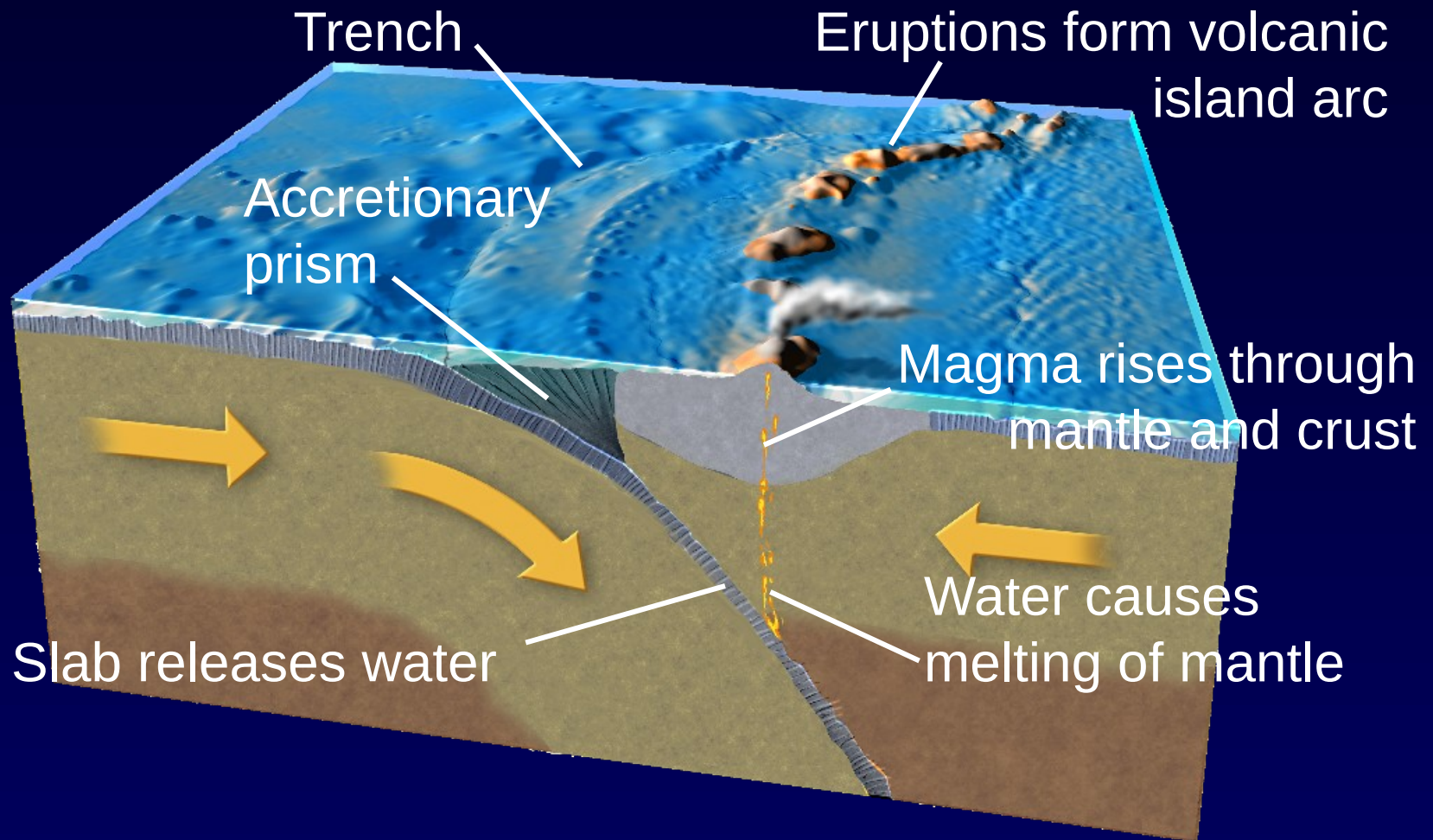
Two oceanic plates
move toward one
another

Trench and island arc



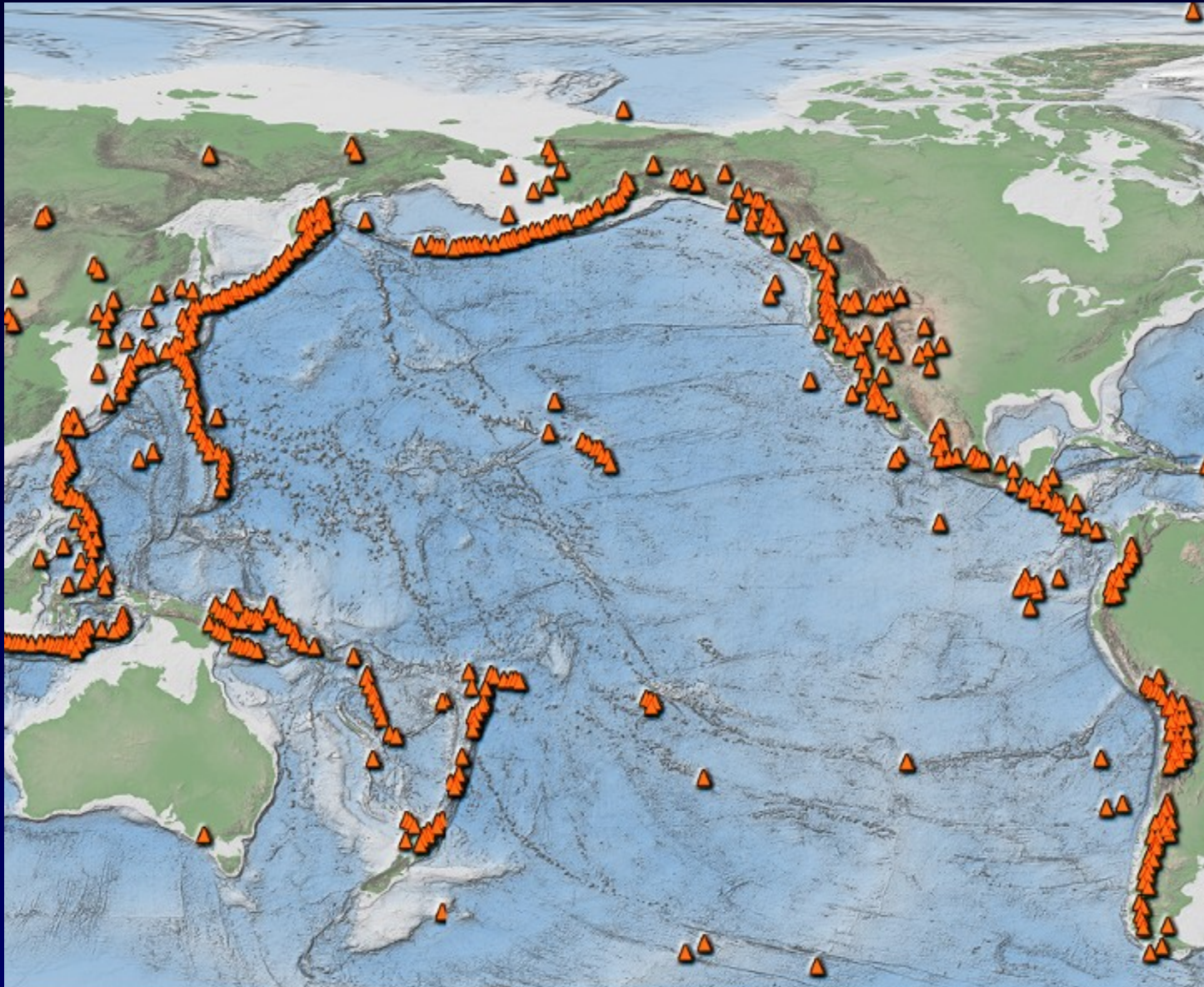
One plate moves
down = *subduction*

Features and Processes in Ocean-Ocean Convergence



Sketch ocean-ocean convergence, labeling the processes in your own words

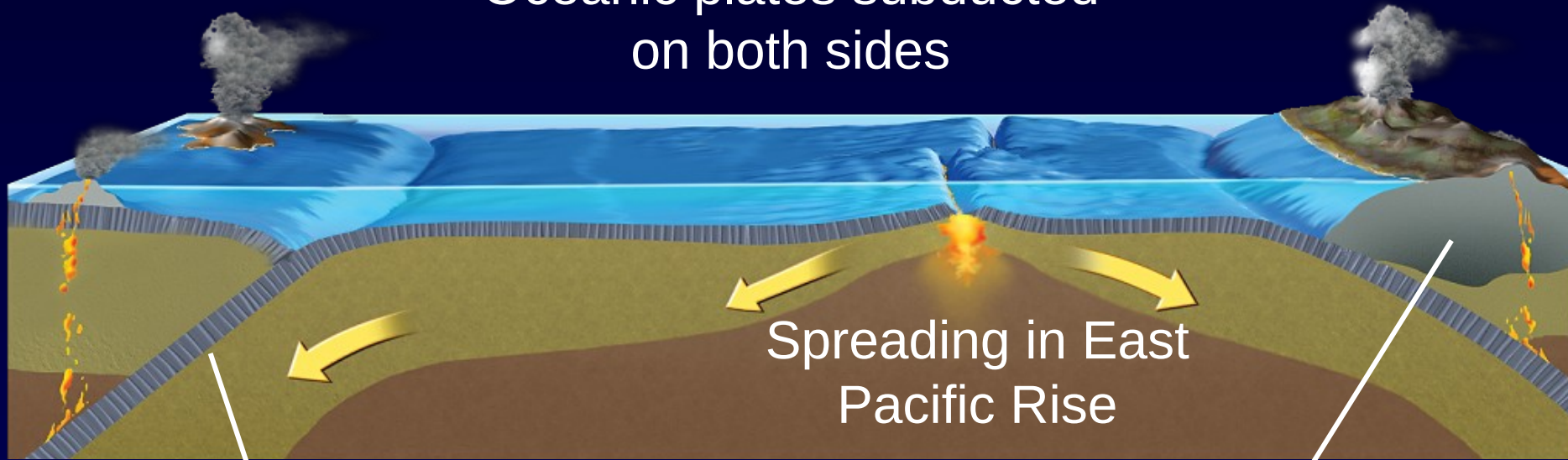
Observe the distribution of volcanoes around the Pacific Ring of Fire



What do you think could explain the overall pattern?

Pacific Ring of Fire

Oceanic plates subducted
on both sides



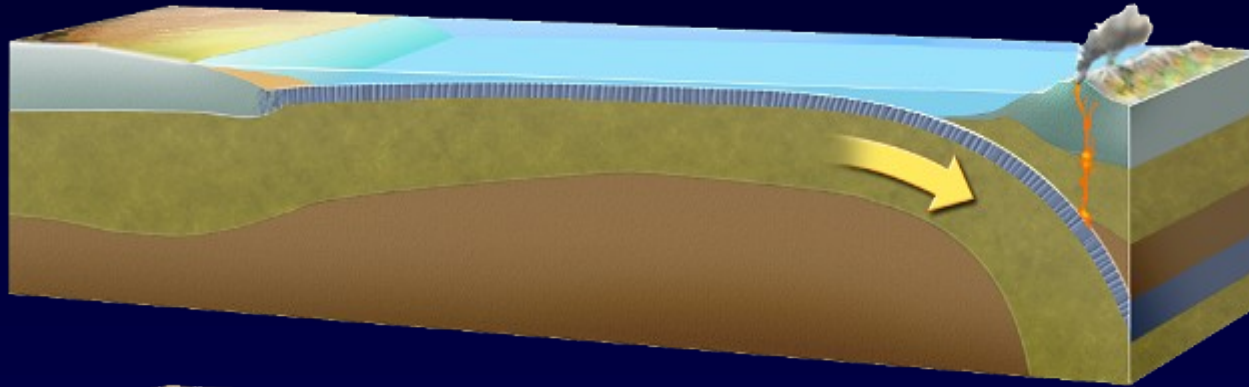
Subduction beneath
oceanic plates = island
arcs (e.g., Japan)

Subduction beneath continental
plates = mountain belts with
volcanoes (e.g., Andes)

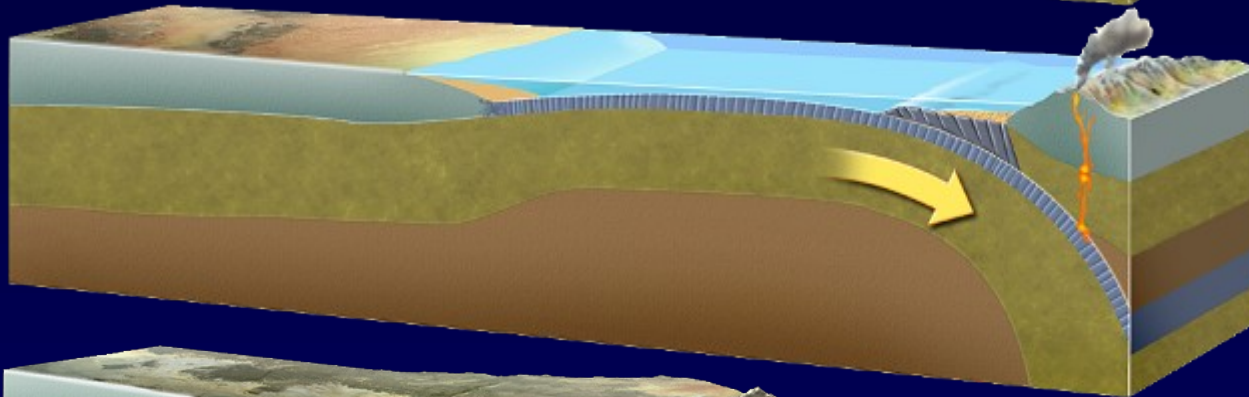
*Explain the Pacific Ring of Fire, including why
the west and east sides are different*

Continent-Continent Convergence

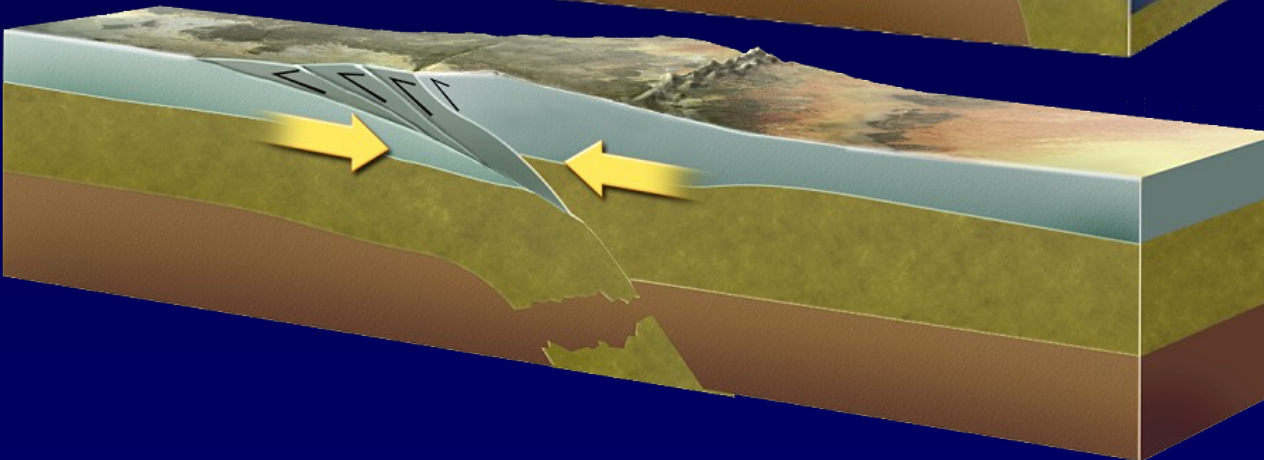
Two continents collide



Subduction
of oceanic
part of plate

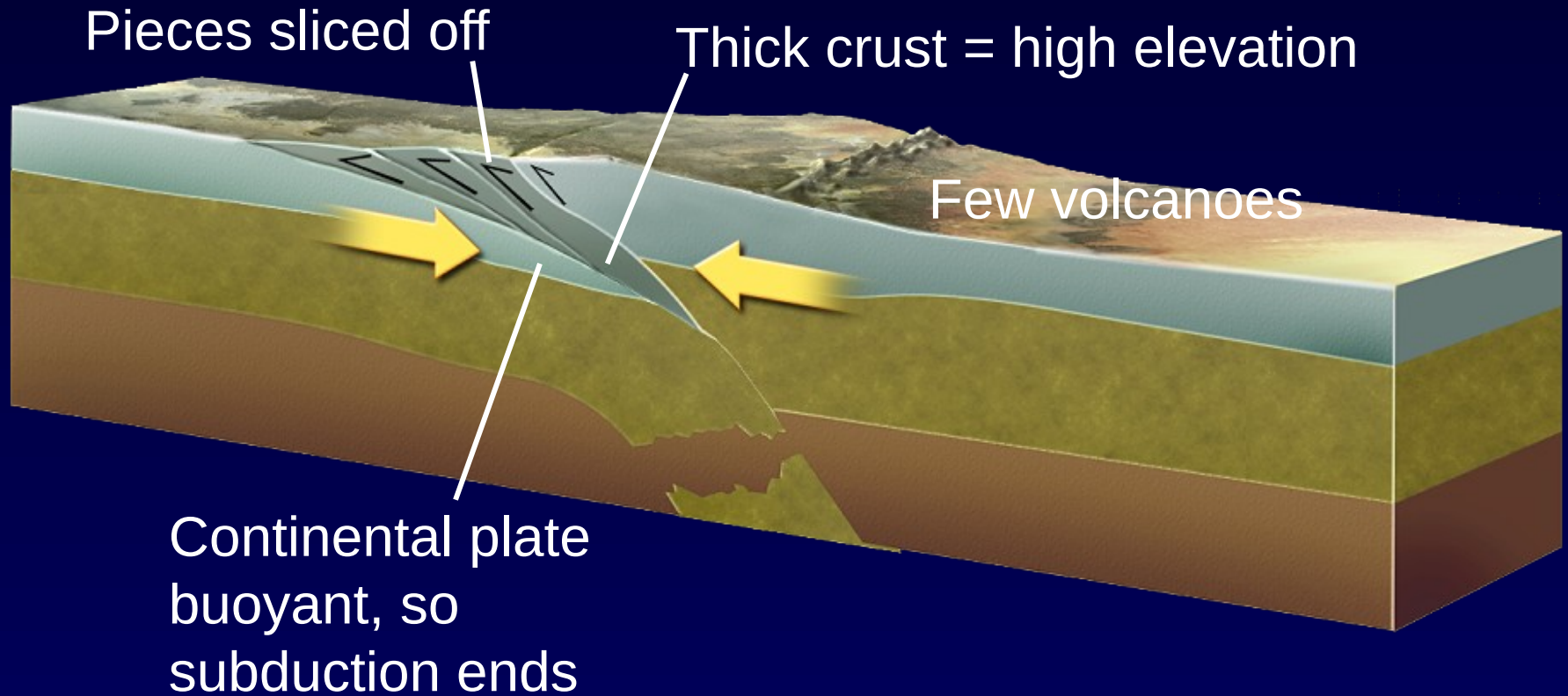


Subduction
brings continents
closer



Continents
collide

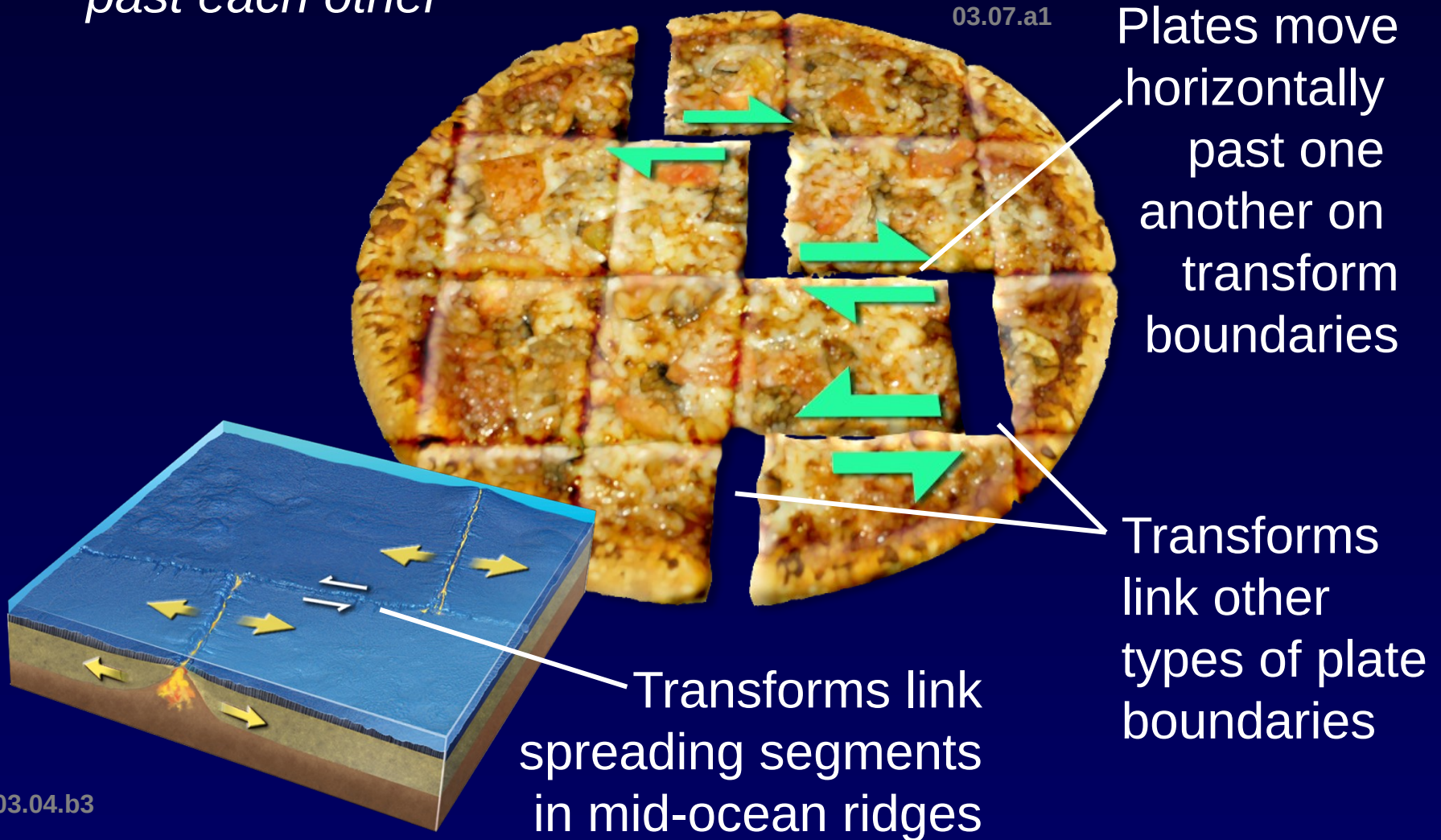
Continental collision = wide zone of deformation

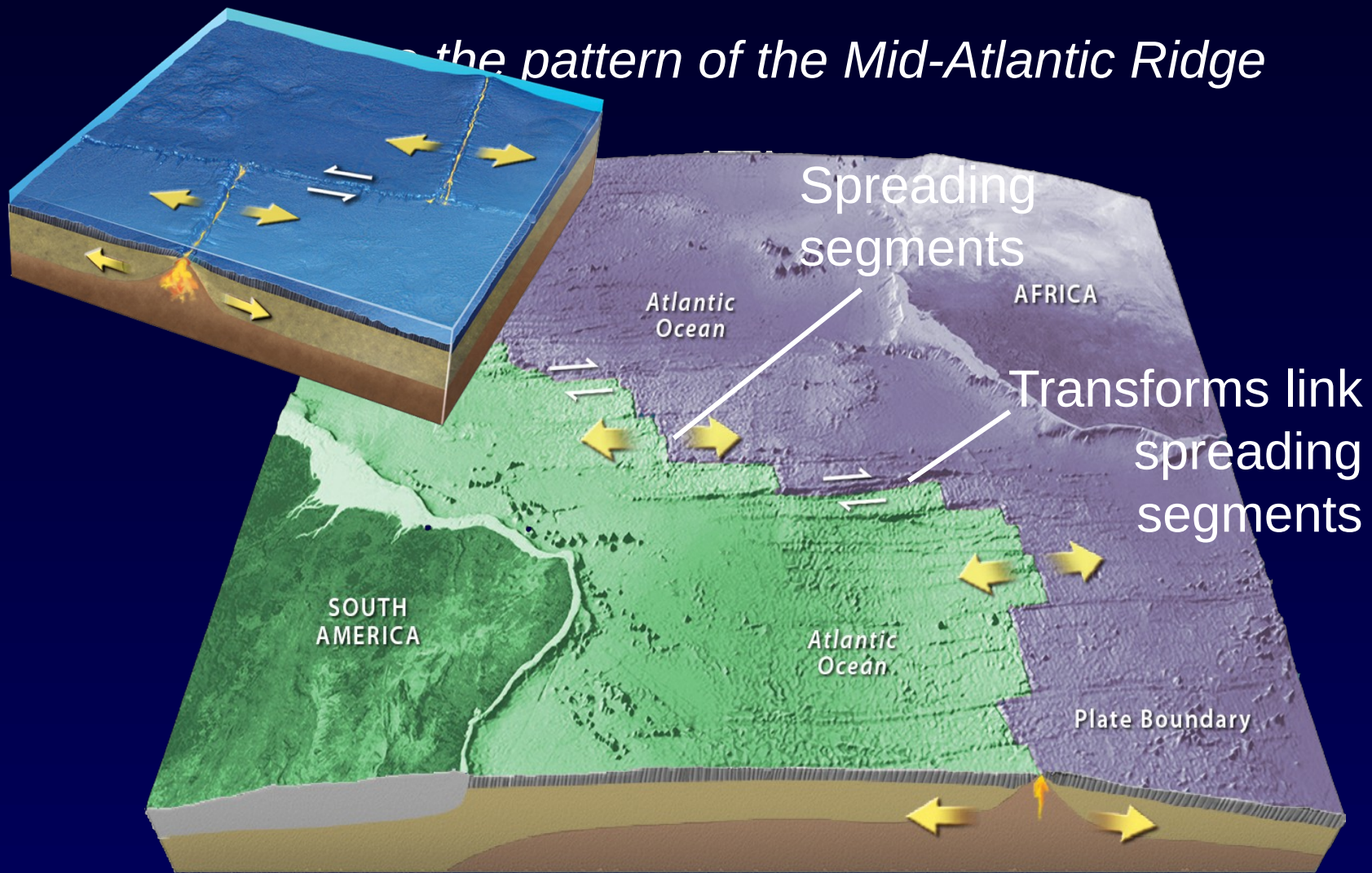


Sketch a continental collision, labeling the processes in your own words

Transform Boundary

Observe how these two “plates” are moving past each other

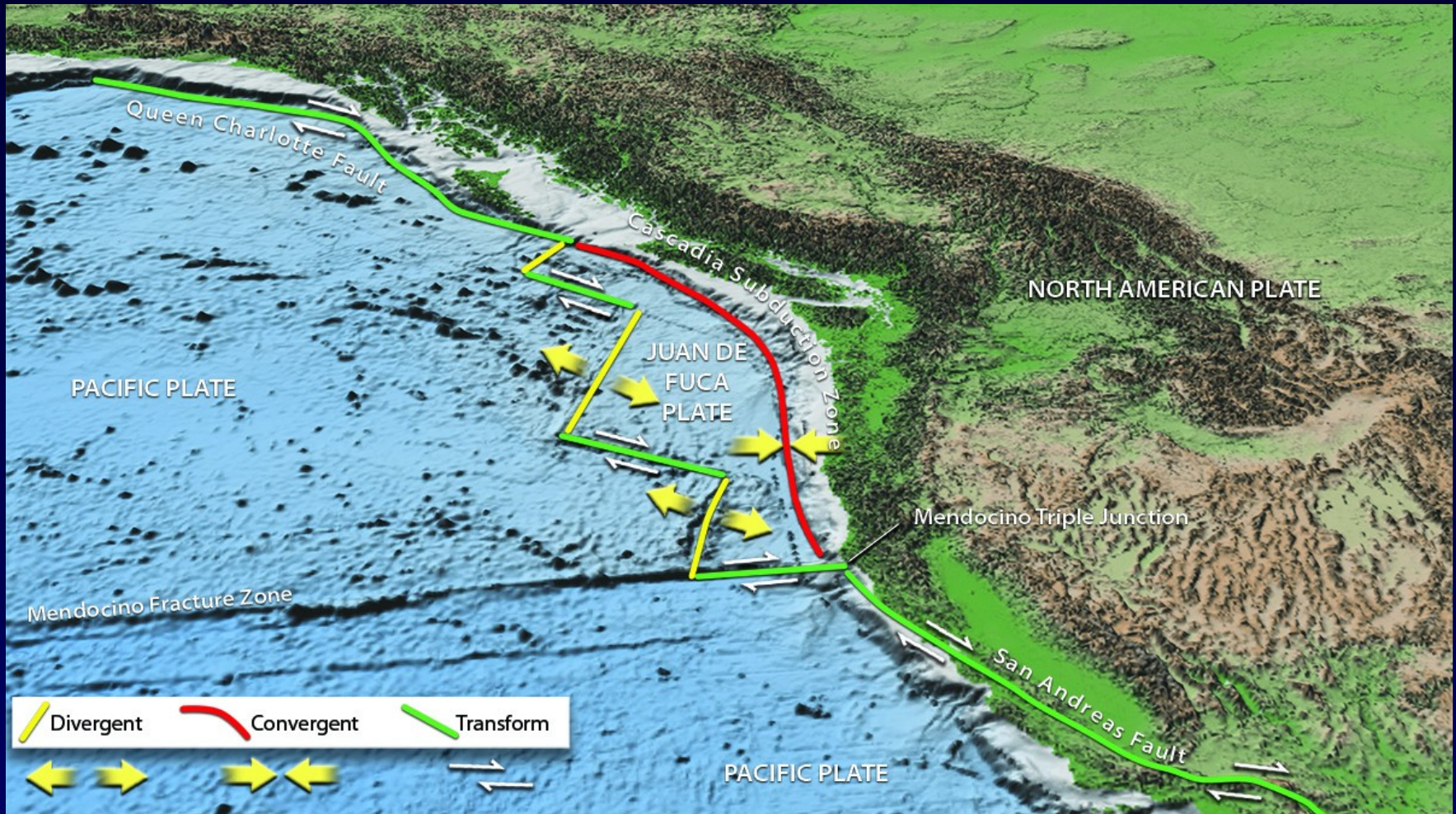




03.07.a2

Sketch a transform boundary, labeling the processes in your own words

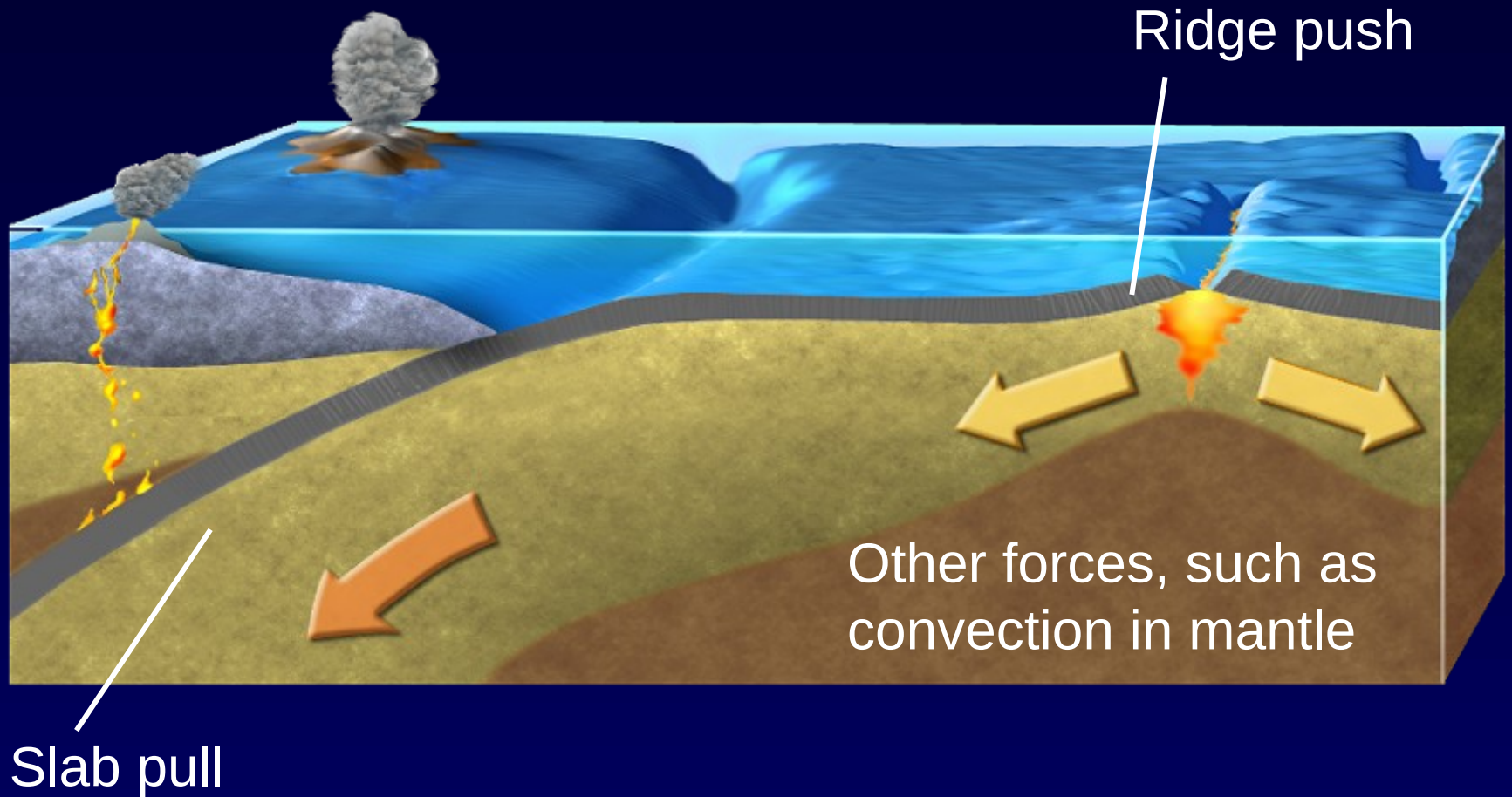
Observe plate boundaries near the west coast of North America (green lines are transform boundaries)



03.07.b1



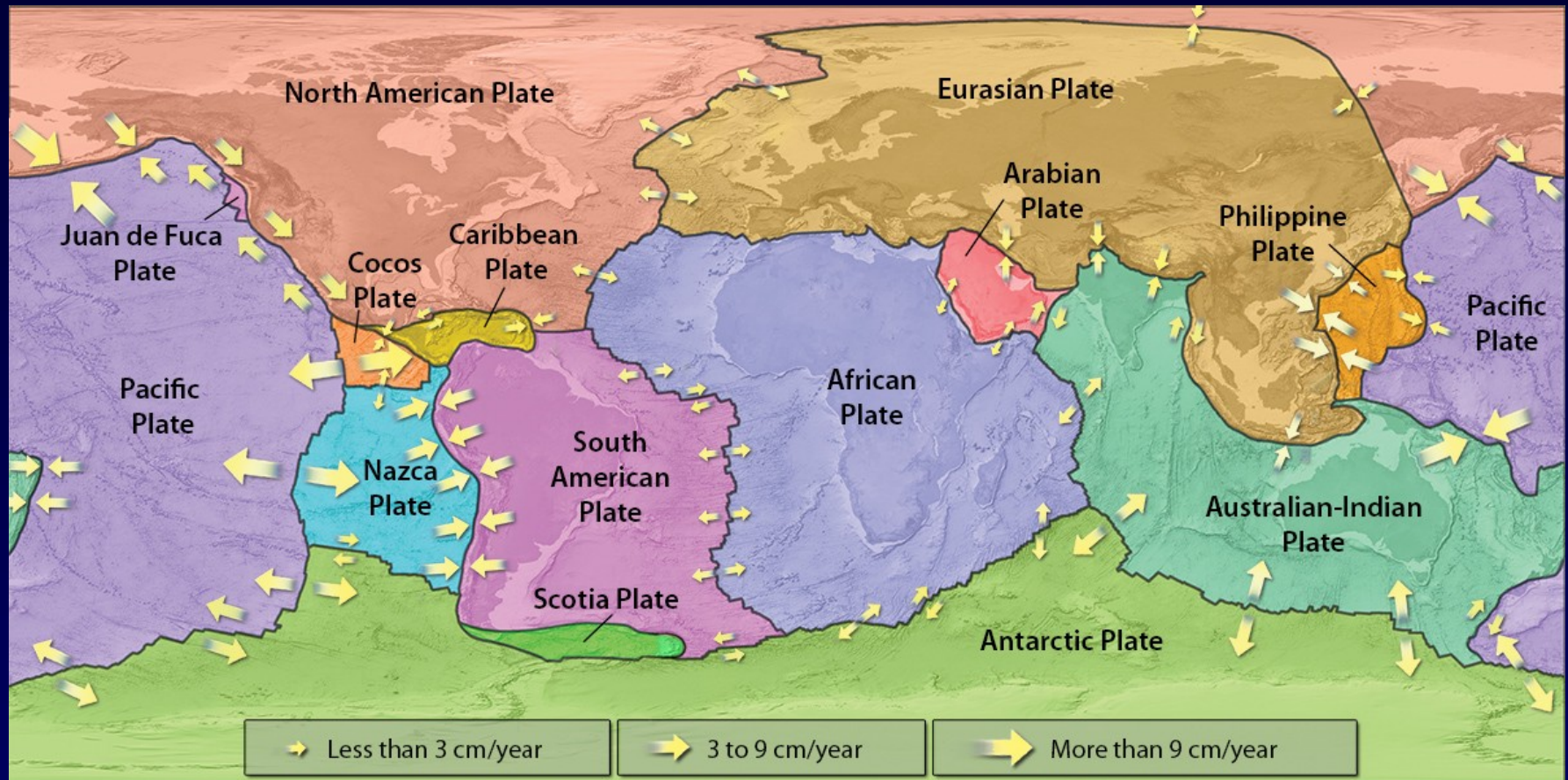
What Moves the Plates?



Rates of Relative Plate Movement

Plates move cm/year

Some move faster than others



03.08.b1

Which plate boundaries have the fastest rates?

Direct Measurements of Plate Motions

Arrows show direction of motion, with higher rates shown by longer arrows

Some arrows are curved because plates are rotating on a sphere



03.08.c1

In what directions are the main plates moving, and how are adjacent plates moving relative to each other?

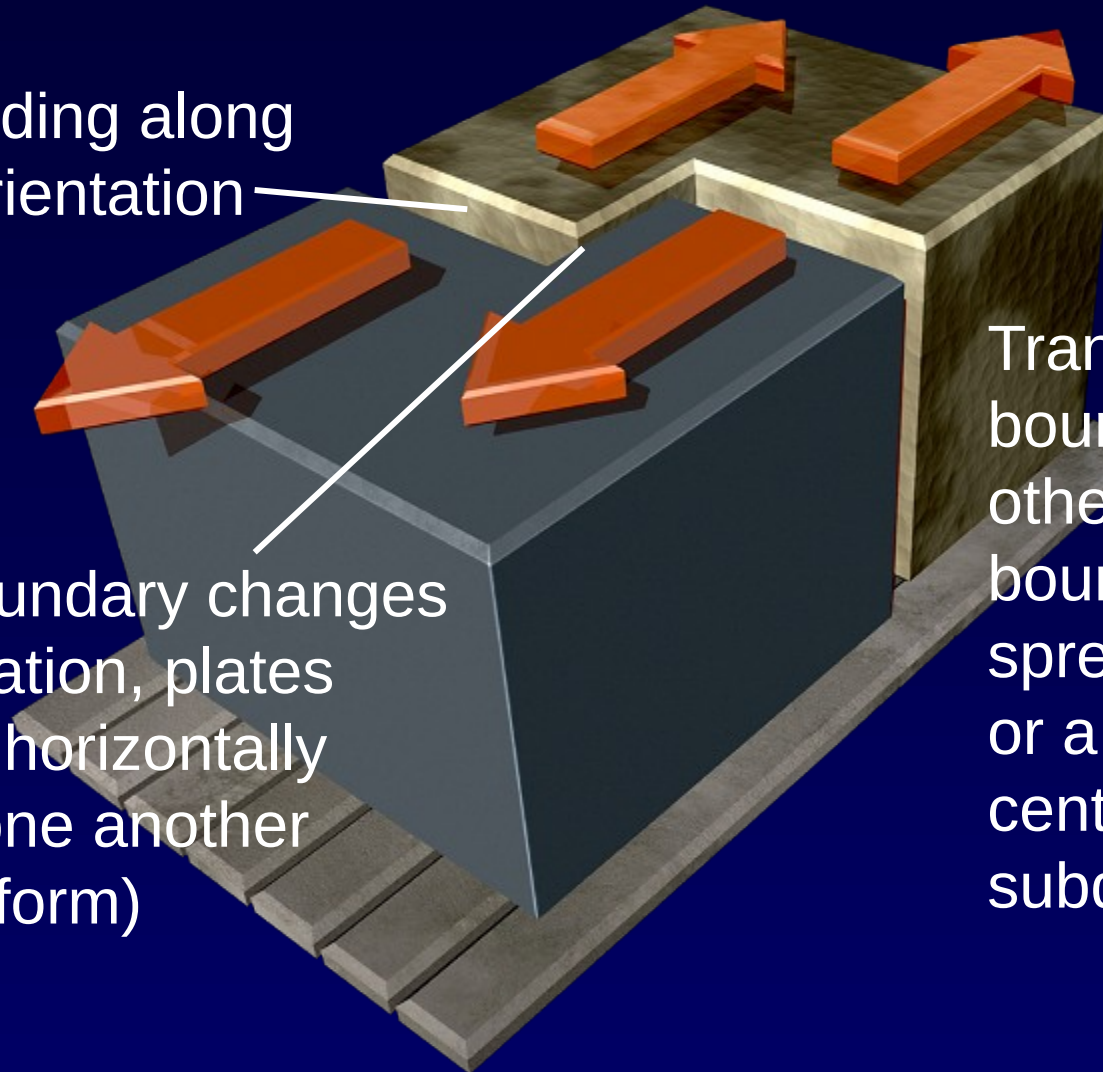
Geometry of Plate Boundaries

Observe how the motion of these two plates varies as the boundary changes orientation

Spreading along
this orientation

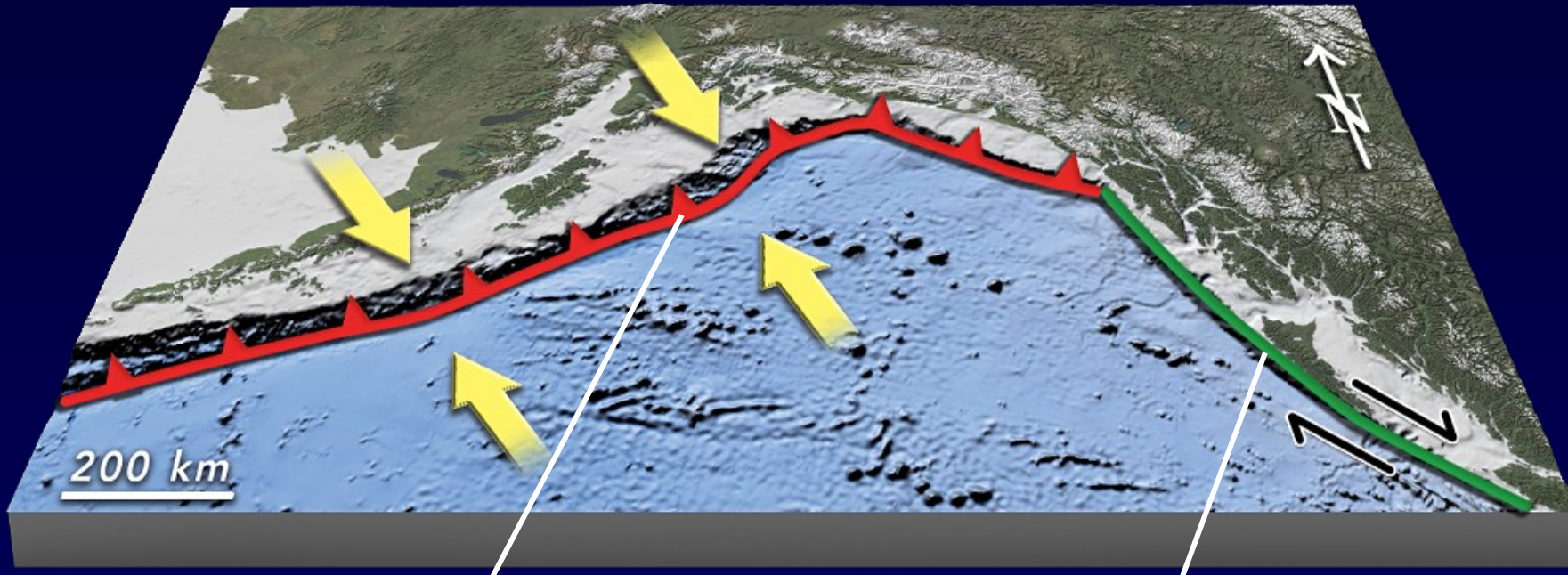
As boundary changes
orientation, plates
move horizontally
past one another
(transform)

Transform
boundaries link
other types of plate
boundaries, like two
spreading centers
or a spreading
center with a
subduction zone



Geometry of Plate Boundaries II

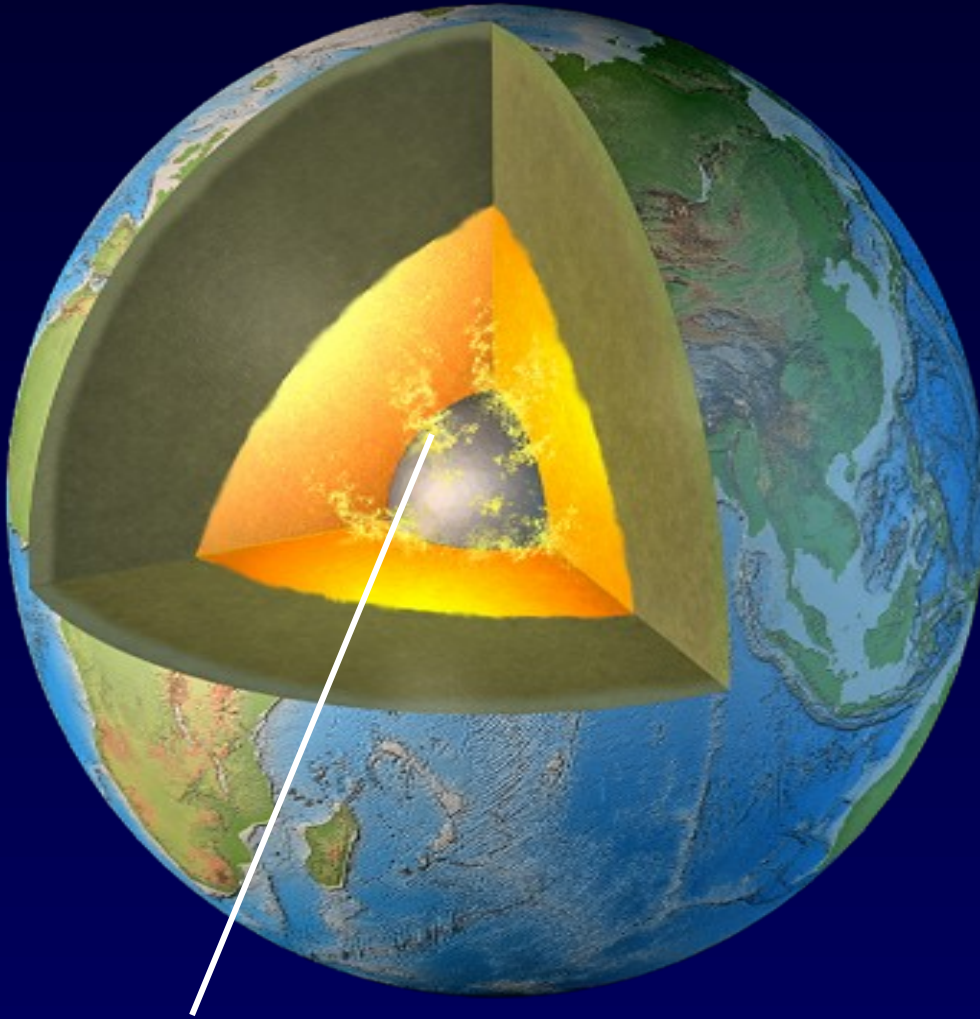
Observe how the motion between the North American and Pacific plates varies as the boundary changes orientation



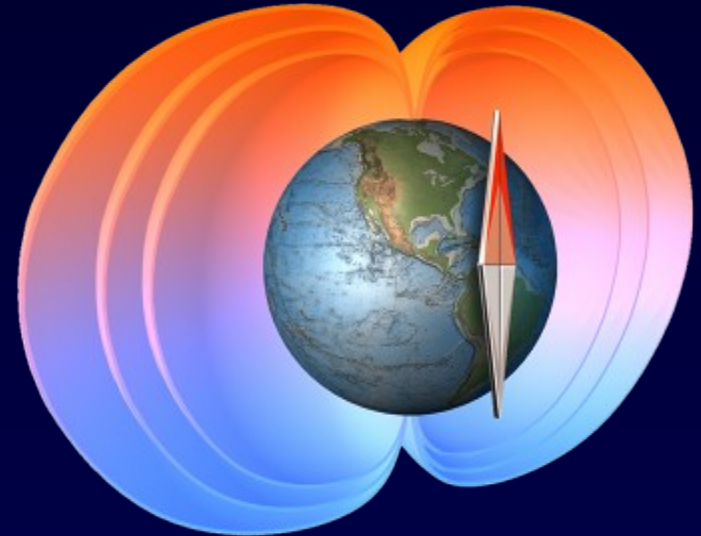
As boundary bends,
becomes convergent
(Pacific plate subducted
beneath NA)

Transform boundary here
(Queen Charlotte fault), with
plates moving horizontally
past one another

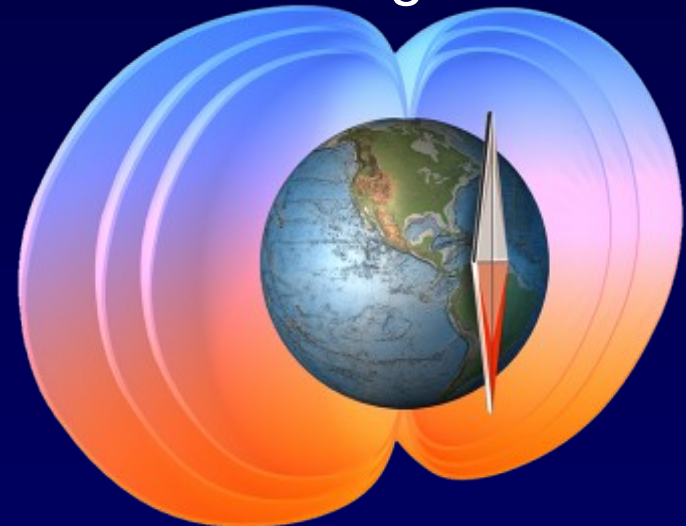
Earth's Magnetic Field: bar magnet S to N



Liquid outer core contains
convection currents



Normal magnetic field



Reversed magnetic field

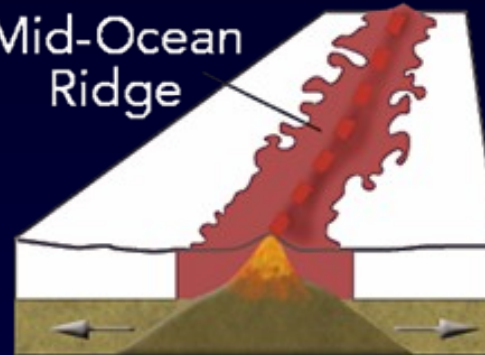
Normal Magnetic
Polarity



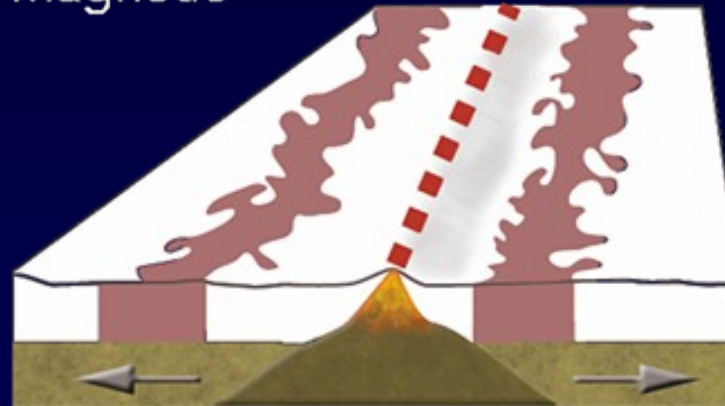
Reversed Magnetic
Polarity



Mid-Ocean
Ridge

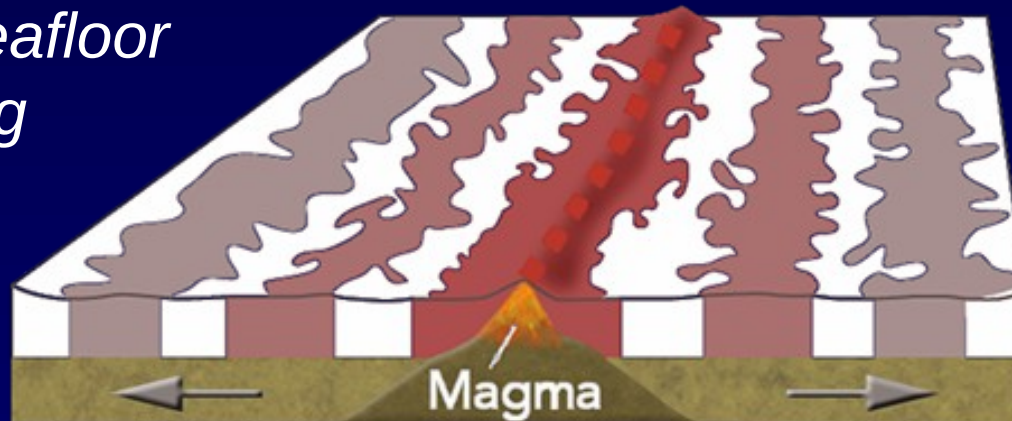


Time 1:
normal
polarity



Time 2:
reversed
polarity;
normal stripe
moved away
from ridge

*Observe how
magnetic
reversals are
recorded by a
mid-ocean ridge
during seafloor
spreading*



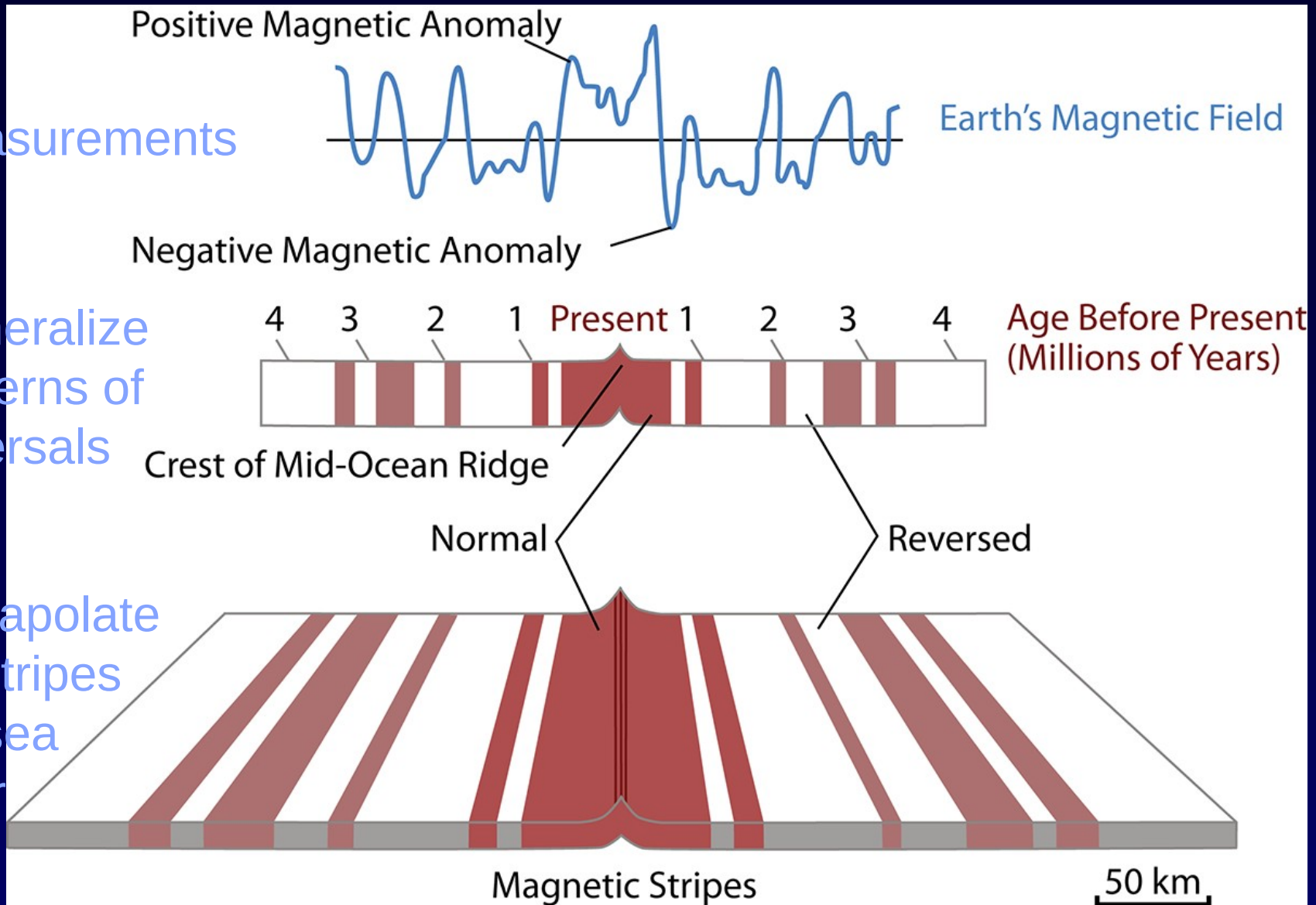
Build series
of magnetic
stripes over
time

Magnetic Stripes on Seafloor

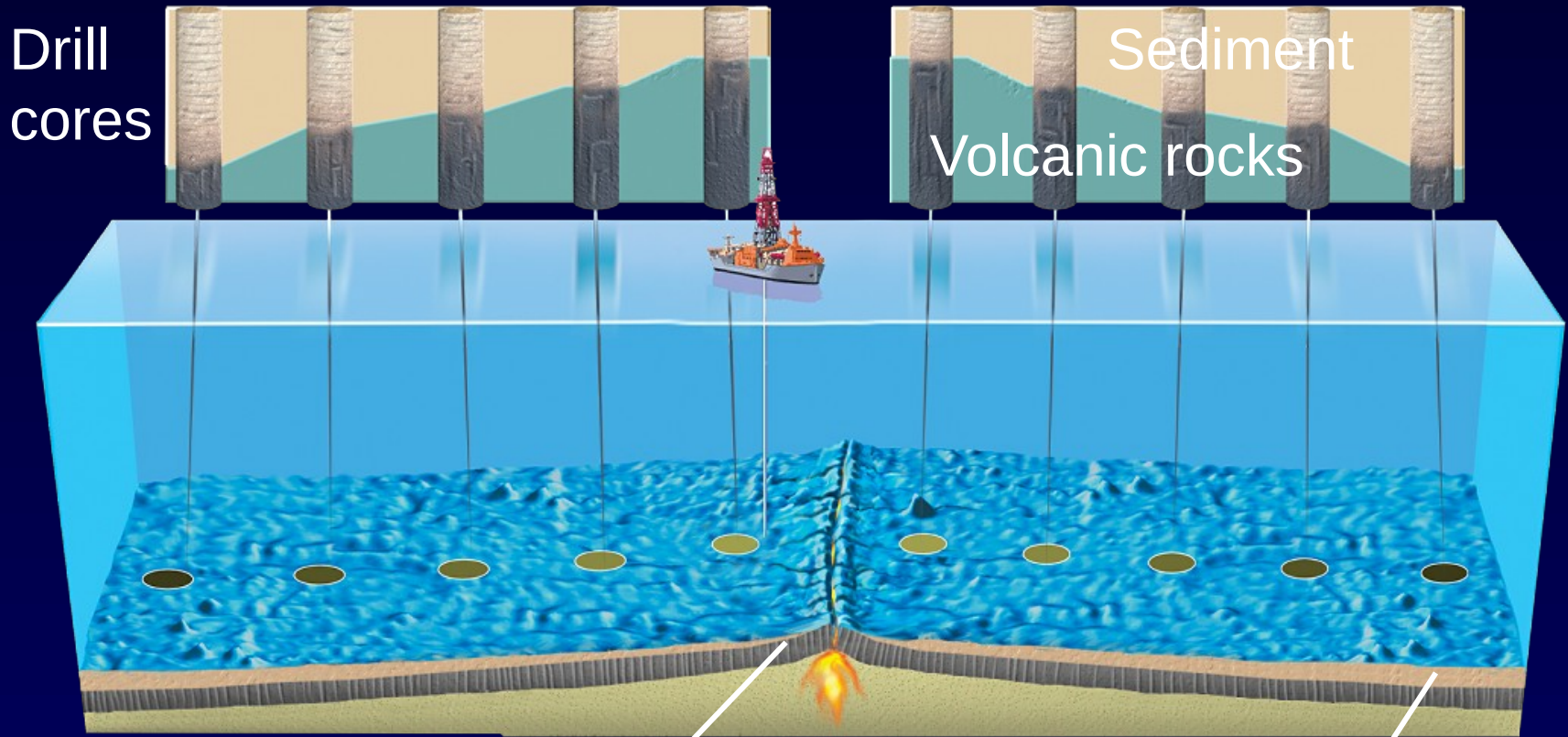
Measurements

Generalize
patterns of
reversals

Extrapolate
as stripes
on sea
floor



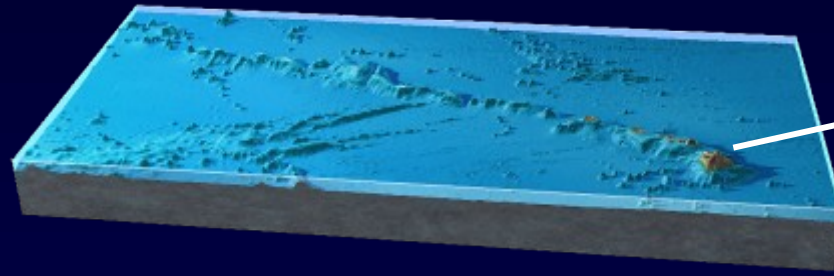
Test of Plate Tectonics: Age of Seafloor and Thickness of Sediment Away from Ridge



Volcanic rocks in oceanic crust are youngest near ridge (just formed)

Sediment thickens away from ridge (had more time to accumulate)

Formation of Linear Island and Seamount Chains



Lines of islands and seamounts (e.g., Hawaii)

Plate moves over a hot spot

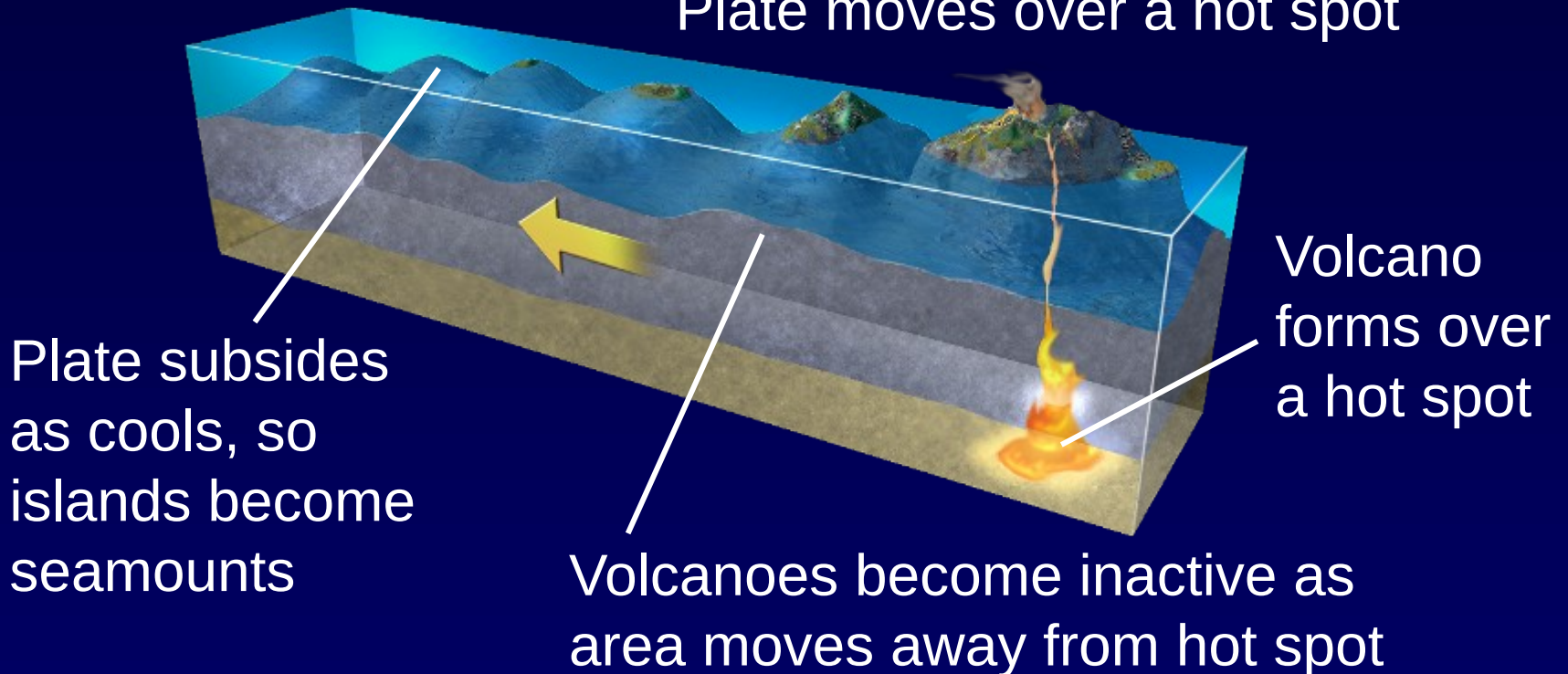


Plate subsides as cools, so islands become seamounts

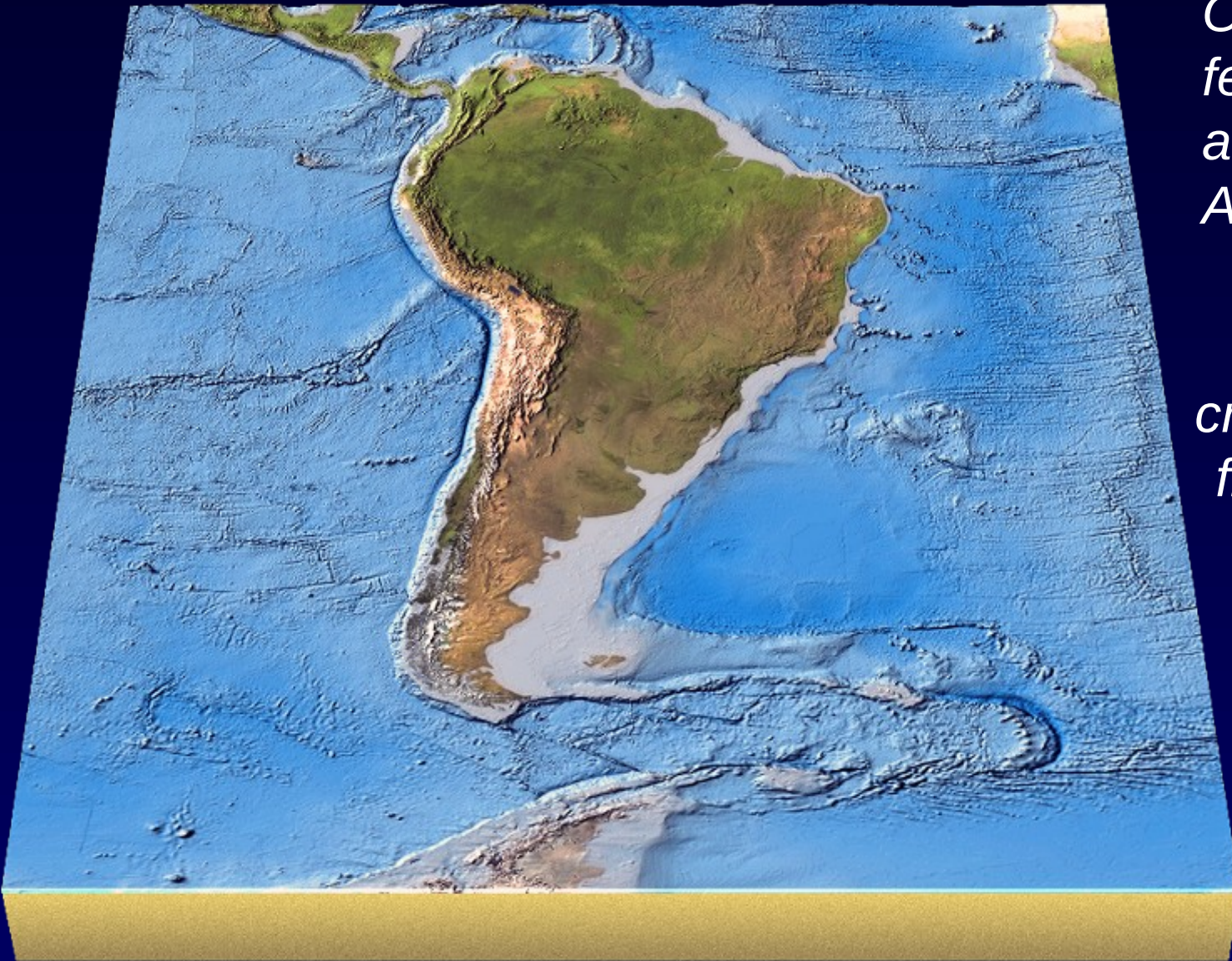
Volcano forms over a hot spot

Volcanoes become inactive as area moves away from hot spot

Why is South America Lopsided

Observe the features around South America

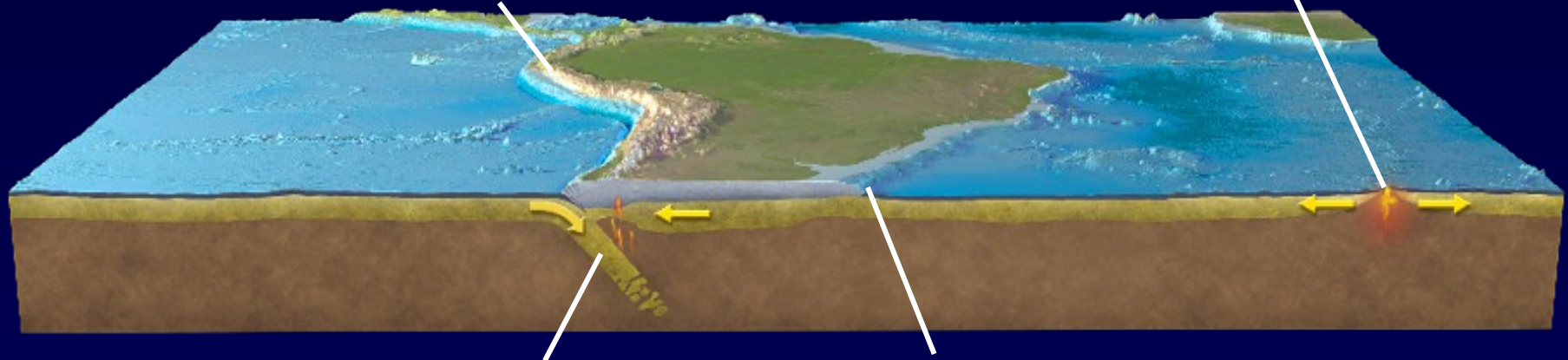
Envision a cross section from west of South America to the Mid-Atlantic Ridge



*Compare this cross section with the
one you envisioned*

Andes (mountains and
volcanoes) over subduction
zone, with trench offshore

Spreading along
mid-ocean ridge



Subduction beneath
western edge

Eastern edge of
continent not a
plate boundary
(passive margin)